Andrology & Sexual Medicine

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · **P2** = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 66	Surgical, Radiographic and Endoscopic Anatomy of the Male Reproductive System
Ch 67	Male Reproductive Physiology
Ch 68	Integrated Men's Health: Androgen Deficiency, Cardiovascular Risk and Metabolic Syndrome
Ch 69	Male Infertility
Ch 70	Surgical Management of Male Infertility
Ch 71	Physiology of Penile Erection and Pathophysiology of Erectile Dysfunction
Ch 72	Evaluation and Management of Erectile Dysfunction
Ch 73	Priapism
Ch 74	Disorders of Male Orgasm and Ejaculation
Ch 75	Surgery for Erectile Dysfunction
Ch 76	Diagnosis and Management of Peyronie Disease
★	Putting it together: andrology decisions

66 Surgical, Radiographic and Endoscopic Anatomy of the Male Reproductive System

1 · SCROTUM

Each layer is a surgical plane from the abdominal wall.

Layer (superficial to deep)	Continuous with	Note
Tunica dartos	Scarpa, Colles, penile dartos	Smooth muscle
External spermatic fascia	External oblique	—
Cremasteric fascia	**Internal oblique**	**Striated** cremaster; thermoregulation
Internal spermatic fascia	Transversalis fascia	—
Tunica vaginalis (parietal, visceral)	Peritoneum	Hydrocele collects between its layers; visceral layer fused to albuginea

- The anterior wall carries the external pudendal vessels and the ilioinguinal and genitofemoral nerves; the posterior wall carries the perineal vessels and nerve. Nothing crosses the raphe.

P1 *Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — scrotum to ipsilateral superficial inguinal; testis to para-aortic/interaortocaval; cauda to external iliac*

> **KEY POINT**
> Scrotal lymph goes to the groin, testicular lymph to the retroperitoneum. †So scrotal violation in testis cancer adds the inguinal nodes to the at-risk field.

2 · TESTIS

Structure

- Volume is **15–25 mL**, length 4.5–5.1 cm. The capsule is visceral tunica vaginalis, **tunica albuginea** (with smooth muscle) and tunica vasculosa.
- The **mediastinum testis** is the posterior echogenic band on ultrasound (US). The appendix testis is a **müllerian** remnant.
- About 250 lobules hold **600–1200 seminiferous tubules**, which drain to the rete testis.
- The interstitium is **20–30%** of volume and holds the **Leydig cells**. **Sertoli** tight junctions form the blood–testis barrier, aided by endothelium and peritubular myoid cells (see Ch 67).

Arterial supply — three sources

Artery	Origin	Point
Testicular (internal spermatic)	Aorta below renal artery	Main supply; larger than the other two combined
Deferential	Superior (or inferior) vesical	Vas, cauda; key collateral, lost after vasectomy
Cremasteric	Inferior epigastric	Tunica vaginalis; anastomoses at mediastinum

- Collaterals may save the testis after testicular artery ligation, but atrophy or azoospermia can occur.

> **KEY POINT**
>
> **Surgical rules**: no traction suture through the lower pole. Biopsy the **medial or lateral upper pole**; the midportion has fewest vessels. After vasectomy the deferential collateral may be gone, so preserve the testicular artery in later cord surgery and counsel about devascularisation.

Veins — pampiniform plexus

The plexus cools the artery by **countercurrent heat exchange** to **2–4 °C below rectal** temperature. A varicocele disrupts this.

Group	Accompanies	Drains to
Anterior	Testicular artery	Inferior vena cava (IVC) on the right; left renal vein on the left
Middle	Vas	Pelvic veins
Posterior	Posterior cord	Epigastric and pudendal veins

P1 *Varicocele – Pathophysiology — left vein into renal vein (**~90% of varicoceles left**), oblique right entry valve-like; middle/posterior groups cause recurrence*

Not in the paper answer:

- **High (retroperitoneal) ligation** can take artery and vein, since collaterals suffice; **inguinal or subinguinal** ligation takes veins only, because arterial ligation there risks devascularisation.

Lymphatics and nerves

- Dividing lymphatics at varicocelectomy causes a **hydrocele**; microsurgery preserves them.
- Supply is autonomic, with **no somatic innervation**; the genital branch of the genitofemoral nerve supplies the tunica vaginalis.
- The **"trifecta" nerve complex** (perivasal, posterior periarterial/lipomatous, intracremasteric) is the target of microsurgical cord denervation for chronic orchialgia.

Imaging

Modality	Role
Ultrasonography (US), 7.5–10 MHz with Doppler	**First line**: pain, mass, swelling, epididymo-orchitis, recurrent varicocele
Computed tomography (CT)	Not primary; cord or inguinal disease; air in Fournier gangrene

Modality	Role
Magnetic resonance imaging (MRI)	Second line: indeterminate US, intra- versus extratesticular; nonpalpable testis (MR angiography)

- If torsion is suspected, **explore**; Doppler must not delay surgery.
- Varicocele is a **clinical diagnosis**. Use US (standing, Valsalva) only if examination is equivocal.

3 · EPIDIDYMIS

- Caput, corpus and cauda lie on the posterolateral testis. **Principal** cells form the blood–epididymis barrier, and **clear** cells acidify the lumen to keep stored sperm quiescent.
- The testicular artery supplies caput and corpus, and the deferential artery the cauda. Nerves and smooth muscle increase toward the cauda, which shifts from storage to propulsion.
- Wolffian (mesonephric) remnants include the **appendix epididymis**; müllerian (paramesonephric) remnants are the **appendix testis** and prostatic utricle.

4 · VAS DEFERENS

- It runs **posterior** to the cord vessels through the inguinal canal, crosses **anterior to the ureter**, forms the ampulla and joins the seminal vesicle.
- It is **30–35 cm** long with a **0.2–0.7 mm** lumen and has **three muscle layers** (longitudinal, circular, longitudinal).
- About half the cord nerves surround the vas: sympathetic fibres drive emission, sensory fibres cause chronic scrotal pain.
- A **vasogram** can cause obstruction, so do it **only at reconstruction**.

5 · SEMINAL VESICLES AND EJACULATORY DUCTS

- Behind the bladder base, they join the ampulla as ejaculatory ducts opening on the verumontanum at **5 and 7 o'clock**.
- They bud from the mesonephric duct alongside the ureteric bud, so **renal anomalies coexist**.
- The fluid is ~70% of semen and alkaline. It carries **fructose**, prostaglandins and **semenogelin**, which forms the coagulum later liquefied by prostate-specific antigen (PSA).
- On **transrectal ultrasound (TRUS)**, a post-ejaculation anteroposterior (AP) diameter **>16 mm** or volume **>6 mL** means obstruction. **MRI** is the modality of choice; CT is for stones.

> **KEY POINT**
> A normal vesicle empties at ejaculation; one that stays large is the imaging signature of ejaculatory duct obstruction.

6 · PROSTATE

P1 *Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer — McNeal zones (peripheral **70%** , transition gives BPH), fasciae, arteries, DVC, lymph, cavernous nerves*

Not in the paper answer:

- TRUS volume is length × width × height × **0.52**, which underestimates. The normal gland is **~25 mL**.
- **Multiparametric MRI** is scored by PI-RADS v2.1; targeted biopsy finds less insignificant cancer.

7 · URETHRA

P1 *Male urethra: embryology and anatomy, endoscopic stricture management, congenital anomalies and epispadias — segments (membranous **2–2.5 cm** ; Cowper 3/9 o'clock), epithelium, sphincter (Onuf), blood supply*

8 · PENIS

Structure

- There are paired **corpora cavernosa** (crura on the pubic arch) and a **corpus spongiosum** (bulb, urethra, glans).
- The septum is incomplete distally, so the corpora act as **one unit** (†one-sided injection fills both).
- The **tunica albuginea** (2–3 mm) has an inner **circular** and an outer **longitudinal** layer; it is thin ventrally.
- **Buck fascia** encloses the corpora and splits around the spongiosum. From midline outward it holds the **deep dorsal vein, then arteries, then nerves**.
- Sinusoidal smooth muscle is **~45%** of volume. Endothelial nitric oxide synthase (NOS) sustains erection.

Arteries

The internal pudendal artery runs through Alcock canal to become the **common penile artery**:

Branch	Supplies
Bulbourethral	Spongiosum, urethra, glans
Cavernosal	Enters at hilum, runs centrally; helicine arteries fill the sinusoids (erection)
Dorsal	Under Buck fascia to the glans; circumflex branches

Veins, nerves

- The superficial dorsal vein drains skin; the **deep dorsal vein** (under Buck fascia) drains to the periprostatic plexus.
- In the distal two-thirds, sinusoids drain via the subtunical plexus and **emissary veins**; the proximal third drains via cavernous and crural veins.
- The **dorsal nerve** (pudendal, **S2–S4**) gives sensation; the **cavernous nerves** (pelvic plexus) give erection.

> **KEY POINT**
> Erection is veno-occlusive: filled sinusoids compress subtunical and emissary veins against the inelastic tunica. †Tunical or smooth-muscle damage defeats this and causes venous leak.

Penile duplex (after intracavernosal injection)

Parameter	Threshold	Meaning
Peak systolic velocity	**<30 cm/s**	Arterial insufficiency
End-diastolic velocity	**>5 cm/s**	Veno-occlusive dysfunction
Resistive index	**<0.80**	Veno-occlusive dysfunction

MRI shows fracture, prosthesis malposition and tumour.

67 Male Reproductive Physiology

1 · HYPOTHALAMIC–PITUITARY–GONADAL AXIS

Every hormonal work-up reads this axis.

	Peptide hormones	Steroid hormones
Examples	Luteinising hormone (LH), follicle-stimulating hormone (FSH)	Testosterone, oestradiol
Storage	Secretory granules	Not stored; secretion equals production

	Peptide hormones	Steroid hormones
Receptor	Cell surface, acting via cyclic AMP (cAMP)	Intracellular, altering transcription; carried bound in plasma

Hypothalamus

- Gonadotrophin-releasing hormone (GnRH) is a decapeptide whose only function is LH/FSH release. Half-life is **5–7 min**.
- Its rhythms are pulsatile (every **90–120 min**), circadian (testosterone peaks early morning) and seasonal.
- **Pulsatility is essential**: a continuous GnRH agonist (leuprolide) desensitises the pituitary, giving castrate testosterone.
- In **Kallmann syndrome** GnRH neurones fail to migrate, which causes hypogonadotrophic hypogonadism. †Anosmia coexists because the neurones migrate with the olfactory placode.

GnRH modulation	Factors
Negative	Opioids (β-endorphin), FSH, LH, testosterone
Positive	Prostaglandin E2, insulin, **kisspeptin** (puberty), **leptin**
Variable	Dopamine (catecholamines)

Anterior pituitary

- LH and FSH share an **α subunit** with TSH and hCG and have a specific **β subunit**.
- **Hyperprolactinaemia abolishes gonadotrophin pulsatility**, so testosterone falls. Low prolactin is not pathological.

	LH	FSH
Pulses	8–16/24 h; follow GnRH	Smaller; partly GnRH-independent (inhibin, activin)
Target	**Leydig cell**	**Sertoli cell** (and spermatogonia)
Action	Steroidogenesis: cholesterol to pregnenolone to testosterone	Tubule growth; **initiates spermatogenesis at puberty**; maintains quantitative output
Main feedback	Testosterone (hypothalamus)	Oestradiol (pituitary) and **inhibin B**

Testicular hormones and feedback

- Testosterone output is ~**5 mg/day**. 5α-reductase converts it to the more potent **dihydrotestosterone (DHT**), needed in most peripheral tissues; aromatase converts it to **oestradiol**, which feeds back especially on FSH.
- FSH makes Sertoli cells produce androgen-binding protein (ABP), transferrin, lactate and inhibin. **Inhibin** suppresses FSH; **activin** stimulates it.

> **KEY POINT**
>
> There are two loops: LH with Leydig cells (testosterone), and FSH with Sertoli cells (inhibin, oestradiol). So **LH reports Leydig function, while FSH and inhibin B report the tubules**. High FSH with low inhibin B means spermatogenic failure.

Development and ageing

- **SRY** (Yp) starts the testis. **DAX1** and **WNT4** are "antitestis" genes; the female pathway is not a passive default.

Fetal cell	Product	Effect
Leydig	Testosterone	Wolffian duct forms internal genitalia
Leydig	Insulin-like peptide 3 (INSL3)	Transabdominal testicular descent

Fetal cell	Product	Effect
Target tissue	DHT	External genitalia
Sertoli	Müllerian-inhibiting substance (MIS/AMH)	Müllerian regression; keeps germ cells out of meiosis

- Puberty (~12 years) is triggered by kisspeptin, leptin and melatonin.
- **Ageing**: testosterone is ~one-third lower by the 7th decade and FSH ~threefold higher. SHBG rises and feedback is blunted, so failure is both peripheral and central.

2 · TESTIS

Architecture and blood flow

- It has **600–1200 tubules** and ~**700 million Leydig cells** in young adults.
- The cord holds one artery in ~half of men, so at microsurgical varicocelectomy ligating a single artery can cause atrophy.
- **Countercurrent exchange** cools arterial blood **2–4 °C** below core and recycles testosterone. It is lost in varicocele and cryptorchidism.

Leydig cells and testosterone synthesis

- **Rate-limiting step**: LH raises cAMP, and **steroidogenic acute regulatory (StAR) protein** moves cholesterol into the inner mitochondrial membrane to become pregnenolone.
- Further enzymes (3β-HSD, 17α-hydroxylase/17,20-lyase, 17β-HSD) make testosterone; their mutations cause 46,XY disorders of androgen biosynthesis.
- †In blood, ~44% of testosterone is bound tightly to SHBG, ~54% loosely to albumin and ~2% is free. Free plus albumin-bound is "bioavailable".

Testosterone peak	Timing	Role
Fetal	12–18 weeks	Builds reproductive tract
Neonatal	~2 months	Imprints target tissues
Pubertal	12–17 years	Masculinisation
Adult	Peak in 2nd–3rd decade, then decline	Maintains androgen-dependent organs

Sertoli cells and blood-testis barrier

P3 P1 *Spermatogenesis; Male reproductive immunology — Sertoli nurse functions, ABP (intratubular androgen ~* ***50-fold*** *serum), basal/adluminal compartments, anatomical + functional barrier, no antibodies before puberty, chemotherapy sanctuary*

Not in the paper answer:

- **Inhibin B** marks Sertoli function; Sertoli tight junctions are the body's strongest, yet open to let spermatocytes pass.

3 · SPERMATOGENESIS

A semen test reflects events about two months earlier.

P3 *Spermatogenesis —* ***~64 days*** *(42–76); proliferation (Ad, Ap every 16 days, B), then meiosis, then spermiogenesis (acrosome from Golgi, flagellum from centriole), then spermiation; six stages, helical wave; cytoplasmic bridges*

Not in the paper answer:

- Output is ~ **123 million/day**, about 1000 per heartbeat.
- †Spermatogenesis needs high intratesticular testosterone, so exogenous testosterone suppresses it.

Stem cells

- Failed migration of primordial germ cells causes extragonadal germ cell tumours or Sertoli cell–only testes.
- "Selfish spermatogonia" (FGFR2/FGFR3 mutations) cause paternal-age disorders and spermatocytic seminoma.

Meiosis

	Mitosis	Meiosis
Divisions / daughters	1 / 2 identical	2 / 4 varied, haploid
Synapsis, crossover	None	Prophase I (synaptonemal complex, chiasmata)
Centromere division	Anaphase	**Anaphase II**

- **Defective recombination** occurs in ~half of maturation arrest. Altered crossover causes aneuploidy, so IVF/ICSI offspring have more chromosomal abnormalities.

Genetics

Y microdeletions of the azoospermia factor (AZF) region occur in **15%** of azoospermic and 6–8% of severely oligospermic men, the commonest defined molecular cause.

Deletion	Histology	Sperm retrieval
Complete AZFa	Sertoli cell–only	Highly unlikely
Complete AZFb	Maturation arrest	Highly unlikely
AZFc	Hypospermatogenesis / focal spermatogenesis	**Often found**
Partial AZFa/b	—	Ejaculated sperm possible

Paternal age matters because spermatogonia divide ~23 times a year, raising de novo dominant mutations (achondroplasia, Apert, Marfan), autism, schizophrenia and **DNA fragmentation**.

4 · EPIDIDYMIS

- The 3–4 m duct runs from caput through corpus to cauda. Testicular sperm are **immotile and cannot fertilise**, except by ICSI.
- **Transport** takes **2–12 days**, driven mainly by spontaneous rhythmic contraction.
- **Storage**: the cauda holds about half the sperm; prolonged storage increases DNA fragmentation.
- **Maturation** gives progressive motility and zona binding. **Fertility is reached mostly by the late corpus or early cauda**.

> **KEY POINT**
>
> Pregnancy occurs even after vasoepididymostomy at the efferent ductules, but **the more distal the anastomosis, the better the fertility**. After vasectomy for >15 years, motility is lower after reversal because of epididymal dysfunction.

- The epididymis is **androgen dependent**, mainly on DHT, which explains impaired semen on 5α-reductase inhibitors.
- **Heat** abolishes storage, and **denervation** (spinal cord injury, diabetes) impairs transport.

5 · VAS DEFERENS, SEMINAL VESICLES, EJACULATORY DUCTS

Semen chemistry shows whether these organs are present and open.

- With a **muscle:lumen ratio of ~10:1** the vas is the most muscular hollow viscus.
- The **seminal vesicle** gives ~**80%** of seminal volume in the late fractions. Its fluid is alkaline and carries **fructose** and prostaglandins; **semenogelin I** forms the coagulum.
- The **ejaculatory duct** has no sphincter; reflux is prevented by its **acute insertion angle**.

- Obstruction may be **physical** or **functional** (vesicle myopathy from diabetes or drugs), and static TRUS may not tell them apart.
- An **ejaculate with pH <7.2, low volume and no fructose** means absent or blocked seminal vesicles, as in congenital bilateral absence of the vas (CBAVD) or ejaculatory duct obstruction.

6 · SPERMATOZOON, CAPACITATION AND FERTILISATION

Part	Contents
Head (4.5 × 3 µm)	Compacted nucleus; acrosome (enzymes)
Neck	Proximal centriole; axoneme starts
Midpiece	~75 mitochondria in a helix, outer dense fibres; ends at annulus
Principal piece / endpiece	Axoneme with fibrous sheath

- The axoneme is **"9 + 2"**, and **dynein** slides the doublets. Inner dynein arm mutants have no motility; outer arm mutants have reduced motility.
- **Paternal mitochondria are degraded** in the egg, so mitochondrial inheritance is maternal.
- **Capacitation** in the female tract gives hyperactivated motility and readiness for the **acrosome reaction**.
- Binding to zona **ZP3** triggers the acrosome reaction. Sperm **phospholipase C zeta (PLCζ)** triggers calcium oscillations, which activate the oocyte.

Sperm defects

- Morphology is graded by strict (Kruger) criteria. Head defects rarely reflect DNA quality, except **pinhead** and **multiple-tailed** sperm.
- In **globozoospermia** (round head, no acrosome) fertilisation fails even with ICSI, because of PLCζ deficiency.

	Ciliary dyskinesia (immotile cilia)	MMAF (multiple morphological abnormalities of the flagella; fibrous sheath dysplasia)
Motility	<10%	Severely reduced
Morphology	Normal on light microscopy; axonemal defect on electron microscopy (EM)	Short, bent, coiled or absent flagella
Cause	Recessive; respiratory disease, dextrocardia (Kartagener)	Genetic; also varicocele, reactive oxygen species (ROS)
Treatment	IVF-ICSI	Correct cause; else IVF-ICSI

7 · EJACULATION PHYSIOLOGY †

Each cause of ejaculatory failure hits a specific phase.

ALGORITHM

1. **Emission** (sympathetic, T10–L2 via the hypogastric nerve; α1): vas, seminal vesicle and prostate contract and deliver semen to the posterior urethra.
2. The **bladder neck closes**. If it fails to close, then ejaculation is retrograde.
3. **Expulsion** (somatic, S2–S4 pudendal): rhythmic bulbospongiosus and pelvic floor contractions, with a relaxed external sphincter, give an antegrade ejaculate.

P3 *Retrograde ejaculation — failed bladder neck closure (α-blocker, diabetes, bladder neck surgery, RPLND) gives retrograde; loss of emission gives anejaculation*

68 Integrated Men's Health: Androgen Deficiency, Cardiovascular Risk and Metabolic Syndrome

1 · THE MEN'S HEALTH CENTRE

The urologist is often the only doctor a man sees.

- Men avoid preventive care but seek help for erectile dysfunction (ED), infertility and voiding, so the urologist is the **front door** to a multidisciplinary centre and triages the whole man.

Age band	Dominant need	Key partners
18–39	Sexual, reproductive, testicular ("rule out hypogonadism"); preventive	Sports medicine, physiotherapy, mental health
40–64	Prostate and colon screening; disease management	Gastroenterology (GI), cardiology, endocrinology, bariatrics, pulmonology
65+	Falls, neurodegeneration, hearing/vision, nutrition	Geriatrics, neurology, audiology, ophthalmology

2 · THE LONGEVITY GAP

- Male life expectancy was **76.1** versus 81.1 years for females (2016).
- Males have higher mortality in **12 of 15** leading causes; the largest gap is **suicide (>3-fold)**.
- Prostate cancer is the only US malignancy with rising stage IV at diagnosis, so discuss screening.

3 · UROLOGIC SCREENING

- Screen sexual and voiding complaints together, with PSA and urinalysis per guidelines.

Questionnaire	Screens
Sexual Health Inventory for Men (SHIM)	Erectile function
Androgen Deficiency in the Aging Male (ADAM)	Androgen-deficiency symptoms
International Prostate Symptom Score (IPSS)	Lower urinary tract symptoms (LUTS)

4 · UROLOGIC DISORDERS OF FOCUS

Testosterone deficiency (TD)

P4 *Male menopause (andropause, late-onset hypogonadism) — low T on more than one occasion + symptoms; AUA **<300 ng/dL** on 2 morning samples; LH separates primary/secondary, prolactin if LH low; TRT contraindications; monitor T, haematocrit (**>54%** hold), PSA at 3–6 months then yearly*

Not in the paper answer:

- For haematocrit, the AUA withholds TRT at >50% and the Endocrine Society at >54%; then reduce the dose or phlebotomise.
- **Randomised trials show TRT benefit** for obesity, metabolic syndrome, diabetes, muscle and bone, but **TRT is not a treatment for depression**.
- For men on chronic opioids, the Endocrine Society advises screening for TD and considering TRT. Exclude secondary causes before starting TRT.

> **KEY POINT**
>
> † **Fertility caution**: exogenous T suppresses LH/FSH and intratesticular T, which causes oligo- or azoospermia. Men wanting children should avoid TRT and use clomiphene, human chorionic gonadotrophin (hCG) or aromatase inhibitors (see Ch 69).

† TRT	Points
Contraindications	Near-term fertility wish; breast cancer; untreated prostate cancer; haematocrit >50%; untreated severe obstructive sleep apnoea (OSA); uncontrolled heart failure; MI/stroke within 3–6 months
Target	Mid-normal T (≈450–600 ng/dL)

† TRT	Points
Formulations	Gels, injections, pellets, each with its own monitoring plan

Erectile dysfunction

- ED affects >50% of men aged 40–70.
- **Every man with ED, of any age, is at cardiovascular disease (CVD) risk**. †Penile arteries are small, so ED precedes coronary events by ~2–5 years.
- Treatment steps from lifestyle to PDE5 inhibitors (never with nitrates), then injection, intraurethral or vacuum therapy, then prosthesis, which has the highest satisfaction. Platelet-rich plasma and shock wave therapy remain in trials.

BPH and infertility

- Refer BPH when medication fails, is not tolerated or is not wanted.
- Infertility is no pregnancy after 1 year; a male factor is present in ~half (see Ch 69).

5 · CARDIOVASCULAR PREVENTION

- Only **lowering low-density lipoprotein cholesterol (LDL-C)** benefits primary prevention; raising HDL or lowering triglycerides does not. Screen lipids from age 20.

Statin decision (ACC/AHA)	Action
LDL-C ≥190 mg/dL	High-intensity statin with or without a nonstatin
LDL-C 70–189, 10-year risk ≥**7.5%**	Moderate–high intensity statin
Risk 5–7.5%	Moderate intensity may be started
Statin-intolerant, risk >20%	Ezetimibe or PCSK9 inhibitor

- USPSTF, NICE and ESC start statins at ≥10% risk.
- A **coronary artery calcium (CAC) score** is most useful at **intermediate risk**; zero means <1% atherosclerosis.

Blood pressure (BP) (ACC/AHA 2017)	Systolic	Diastolic
Normal	<120	and <80
Elevated	120–129	and <80
Stage 1	130–139	or 80–89
Stage 2	≥140	or ≥90

Elevated or stage 1 BP gets lifestyle change. Drugs are for stage 2, or stage 1 with age ≥65, T2DM, CVD, chronic kidney disease or 10-year risk ≥10%.

6 · OBESITY AND METABOLIC SYNDROME

Fat lowers testosterone, and low testosterone adds fat.

- Body mass index (BMI): overweight 25–29.9; obesity I 30–34.9, II 35–39.9, III ≥40.
- Advise weight loss at BMI ≥30, or 25–29.9 with CVD risk, OSA or osteoarthritis; GLP-1 agonists are approved.
- †Metabolic syndrome (ATP III) needs ≥3 of: waist >102 cm, triglycerides ≥150 mg/dL, HDL <40 mg/dL, BP ≥130/85, glucose ≥100 mg/dL. Adipose aromatase raises oestradiol, which lowers LH, so low T and metabolic syndrome drive each other.

7 · GI, SKIN AND MENTAL HEALTH

- Evaluate **constipation** in every man with UTI or retention. Colorectal screening runs from **45–75** years.
- Iron deficiency anaemia carries a >30-fold GI malignancy risk and needs investigation.

- Psoriasis is linked to MI, and suicide is much higher in men.

69 Male Infertility

1 · DEFINITION AND OVERVIEW

Half of infertility involves the man, so always examine him.

- Infertility is no pregnancy after **1 year** of unprotected intercourse. It affects ~15% of couples; male and female factors contribute equally.
- Most conceptions occur in the first 3–6 cycles, so evaluation may start after 6 months.
- Idiopathic infertility is the largest group. **Varicocele is the commonest identifiable cause**, then obstruction.

† Classification	Examples
Pre-testicular (endocrine)	Hypogonadotropic hypogonadism (HH), Kallmann, prolactinoma, exogenous androgens, opioids, obesity, congenital adrenal hyperplasia (CAH)
Testicular	Klinefelter, Y microdeletion, cryptorchidism, torsion, orchitis, varicocele, chemotherapy/radiation, Leydig failure
Post-testicular	Obstruction (epididymal, vasal, ejaculatory duct), congenital bilateral absence of the vas (CBAVD), ejaculatory dysfunction, antisperm antibodies, structural sperm defects

2 · HISTORY

P2 *Discuss the role of urologist in reproductive endocrinology and male infertility practice — history, examination, endocrine tests (LH localises, FSH/inhibin B = germ-cell mass, prolactin), HH/hyperprolactinaemia/CAH treatment, surgical roles, ART choice, testis cancer in azoospermia*

Not in the paper answer:

- The **female partner's age is the most important determinant**; fecundity falls after 35.
- Paternal age ≥40 needs counselling about offspring bipolar disorder, schizophrenia, autism and ADHD (ASRM/AUA 2021).
- Semen quality peaks after **1–2 days' abstinence**, so advise daily intercourse around ovulation; most lubricants impair sperm.

Gonadotoxins and drugs

> **KEY POINT**
>
> **Exogenous testosterone (T) is a contraceptive**. It is aromatised to oestradiol (E2), which feeds back on the pituitary and lowers LH. Intratesticular T then falls, and spermatogenesis falls with it. Recovery usually takes ≥4 months after stopping (after ≤2 years' use); after longer use it is unknown.

- Other harmful drugs include antiandrogens, spironolactone, corticosteroids, anabolic steroids and some antiretrovirals. Cannabis, alcohol and smoking also impair semen.

Infection and childhood conditions

- Infection injures either **anatomically** (duct scarring) or **functionally** (leucocytes, reactive oxygen species [ROS]).
- After torsion, about half have later spermatogenic effects.

Cryptorchidism (after orchidopexy)	Unilateral	Bilateral
Paternity	Near normal	Markedly reduced
Inhibin B	~⅔ of controls	~⅓ of controls

- **Laterality is the key prognostic factor.**

- The American Urological Association (AUA) advises orchidopexy within **18 months**, because germ cells are lost from 15–18 months. Later surgery before 10 years still improves sperm retrieval.

3 · PHYSICAL EXAMINATION

Testis size and the vasa sort most azoospermic men.

- Examine the man standing. An engorged epididymis suggests obstruction, and testis size reflects spermatogenic mass.

Testis size	Abnormal
Seager caliper (long axis)	**≤4.6 cm**
Prader orchidometer (volume)	≤20 mL
Scrotal ultrasound (US)	Most accurate; added value unclear

- Unilateral vasal absence needs renal US; bilateral absence needs CFTR testing and renal US.
- Palpable seminal vesicles suggest ejaculatory duct obstruction (EDO).

Varicocele grade	Definition
Subclinical	Seen only on Doppler; **do not treat**
I	Palpable only with Valsalva, standing
II	Palpable without Valsalva, not visible
III	Visible through scrotal skin

4 · ENDOCRINE EVALUATION

ALGORITHM

1. Take a morning sample for total T, sex hormone-binding globulin (SHBG) and albumin, and calculate bioavailable T.
2. Measure LH and follicle-stimulating hormone (FSH), then E2.
3. Measure prolactin only if indicated.

- Adequate total T is ≥**280–300 ng/dL**. Low T is common even with density >20 million/mL, so assess every infertile man.
- Low SHBG (obesity, diabetes) makes total T under-read; high SHBG (ageing, cirrhosis) makes it over-read.
- A T (ng/dL) to E2 (pg/mL) ratio **<10:1** means aromatase excess.
- Prolactin of 20–50 µg/L needs no further work-up; a significant rise needs pituitary MRI.

5 · SEMEN ANALYSIS

P4 *Semen analysis nomenclature — collection, ≥2 samples 2–3 weeks apart; WHO 2021 5th centiles (volume **1.4** mL, **16** million/mL, 39 million, 42/30%, **4** % strict forms, 54% vitality); terms; only azoospermia definitive; HOS test*

Not in the paper answer:

- **Low volume (<1.0 mL)** suggests EDO, CBAVD, hypoandrogenism or retrograde ejaculation. Do a **postejaculatory urinalysis**; if there are no sperm in urine, do transrectal ultrasound (TRUS).
- †Sperm DNA fragmentation testing is not routine; use it for recurrent pregnancy loss or failed assisted reproduction (ART).

6 · GENETIC TESTING

ALGORITHM

1. If non-obstructive azoospermia (NOA) or density **<5 million/mL**, then do karyotype and Y-microdeletion testing (ASRM/AUA 2021).
2. If CBAVD (or idiopathic EDO), then test CFTR in the man **and his partner**.

P3 *Genetic causes of male infertility. Enumerate and classify. — AZFa (USP9Y, DDX3Y)/AZFb complete deletion: no retrieval; AZFc (DAZ): sperm often found, sons inherit; CFTR; Kallmann; androgen receptor; globozoospermia, AURKC, PCD*

Not in the paper answer:

- AZFc deletion with oligozoospermia declines progressively, so **cryopreserve sperm** early.

7 · IMAGING

- Do not screen, because incidental microlithiasis causes harm. Varicocele is clinical.

P3 *Ejacultory duct obstruction — TRUS vesicle AP **>1.5 cm** with or without midline cyst (low specificity); aspiration ≥3 sperm/HPF; vesiculography, chromotubation; <5% of azoospermia; idiopathic EDO CFTR in ~half; TURED with TRUS guidance*

8 · AZOOSPERMIA: OBSTRUCTIVE (OA) VS NON-OBSTRUCTIVE (NOA)

This split decides between reconstruction and sperm extraction.

	OA	NOA (spermatogenic dysfunction)
FSH	**≤7.6 IU/L**	>7.6 IU/L
Testis long axis	>4.6 cm	≤4.6 cm
Management	Reconstruction (see Ch 70) or sperm retrieval	Genetic tests; microdissection testicular sperm extraction (micro-TESE) with intracytoplasmic sperm injection (ICSI)

ALGORITHM

1. If the vasa are absent, then test CFTR in both partners.
2. If the vasa are present and volume is <1 mL, then do postejaculatory urinalysis. Sperm in urine means retrograde ejaculation; no sperm means TRUS for EDO.
3. If volume is normal, then use FSH and testis size to separate OA from NOA.

9 · SPECIFIC CAUSES AND TREATMENT

Klinefelter syndrome (47,XXY)

P4 *Klinefelter's syndrome. — commonest genetic cause; 1 in 500–1000; non-mosaic 80–90%; small firm testes, raised FSH/LH, low T; ejaculated sperm ~8%; micro-TESE ~ **50%** ; mediastinal GCT*

- **Trap**: exogenous T suppresses residual spermatogenesis, so extract sperm before starting T. If fertility is wanted, use aromatase inhibitors or human chorionic gonadotrophin (hCG).

Hypogonadotropic hypogonadism — most treatable cause

State	Hormones	Treatment
Kallmann (anosmia; KAL1, GNRHR)/complete HH	Low LH, low FSH	hCG with or without recombinant FSH (rFSH) or human menopausal gonadotrophin (hMG)
Isolated LH deficiency ("fertile eunuch")	Low LH, normal FSH	hCG
Isolated FSH deficiency	Low FSH	FSH
Hypothalamic GnRH deficiency	Low GnRH	Pulsatile subcutaneous GnRH every 2 h, for ≥6 months

State	Hormones	Treatment
Partial HH (common)	LH inappropriately low for T	Selective oestrogen receptor modulator (SERM) or aromatase inhibitor

- Give hCG 1500–5000 IU 2–3 times weekly (max 10,000/week), titrated to T; add hMG or rFSH 75 IU 2–3 times weekly. After androgen abuse, allow 1–2 years.
- Treat prolactinoma with cabergoline, and CAH with corticosteroids.

Empirical medical therapy

Drug	Dose	Notes
Clomiphene (SERM)	25 mg daily or 50 mg alternate days; max 100 mg	Blocks E2 feedback, so LH and FSH rise; off-label; no change in prostate-specific antigen (PSA) or haematocrit; target T 600–800 ng/dL
Anastrozole / letrozole	Anastrozole 1 mg daily	Raises T and lowers E2; use when T:E2 <10:1 (obesity); caution with bone density

Primary failure and androgen resistance

- High LH with low T has no accepted therapy; exogenous T is contraindicated, so offer sperm extraction.
- Androgen receptor defects give high T, LH and E2 and are treated with ICSI.

CBAVD

P3 *CBAVD — genital form of CF; mild CFTR alleles; normal spermatogenesis, low-volume azoospermia; partner carrier ~1 in 20; renal agenesis ~1 in 10; retrieval + ICSI*

Varicocele

P3 *Varicocele: controversies and present status of treatment — prevalence, left predominance, right-sided: exclude tumour; heat/ROS/DNA injury; repair palpable + abnormal semen, never subclinical; may down-stage ART*

- Varicocele affects ~15–20% of all men versus 35–50% of infertile men (see Ch 70).

Antisperm antibodies and structural defects

- For antisperm antibodies, washing fails and steroid trials conflict, so use **ICSI**.
- Globozoospermia needs ICSI with or without oocyte activation, and ICSI is not recommended for macrocephalic multiflagellate sperm (AURKC).

10 · LIFESTYLE AND OBESITY

- Obesity raises E2 and lowers LH and T, and worsens ART outcomes, so advise weight loss. Stop smoking, cannabis, excess alcohol and anabolic steroids, and avoid heat.

11 · ASSISTED REPRODUCTION

Sperm count decides the technique.

- †Choose by total motile sperm count (TMSC): IUI if processed TMSC is ≳5–10 million; IVF below that or after failed IUI; ICSI for severe oligozoospermia, retrieved sperm, antisperm antibodies or prior fertilisation failure.
- †Retrieval in OA is by percutaneous or microsurgical epididymal sperm aspiration (PESA/MESA) or testicular aspiration; in NOA it is micro-TESE.

70 Surgical Management of Male Infertility

1 · PRINCIPLES

- In vitro fertilisation (IVF) with intracytoplasmic sperm injection (ICSI) treats even non-obstructive azoospermia (NOA), but it is costly, burdens the female partner and bypasses natural genetic barriers.
- Treating the man (reconstruction for obstructive azoospermia (OA), varicocelectomy) is safest and cheapest, and moves the couple **down the ladder**: donor sperm, then IVF/ICSI with testicular sperm, then IVF/ICSI with ejaculated sperm or intrauterine insemination (IUI), then natural conception.

2 · SURGICAL ANATOMY

Structure	Arteries (origin)
Testis	**Testicular** (aorta) — main; deferential (internal iliac/superior vesical); cremasteric (inferior epigastric; mainly tunica vaginalis)
Epididymis	Superior/medial epididymal (from testicular a.) supply the caput; inferior epididymal (from deferential a.) supplies the cauda; interconnected
Vas	Abdominal end: deferential only. Testicular end: deferential and inferior epididymal. **No supply from cremasteric or cord vessels**

- The testicular artery **exceeds deferential and cremasteric combined**; after vasectomy, preserve it at any later surgery.
- The epididymis can be dissected off the testis if the **superior epididymal artery is intact**; this gains **4–6 cm**.
- If the vas is blocked and devascularised at two sites, the segment between fibroses. So **never do two simultaneous vasovasostomies on one vas**; excise it and do one tension-free anastomosis.
- **Excurrent ducts**: 7–11 efferent ducts enter the caput, but a **single tubule** runs from the caput–corpus junction to the vas. So injury distal to the caput (even **one stitch**) obstructs the whole side.

3 · DIAGNOSTIC TESTIS BIOPSY

P1 *Testicular biopsy and TESA (testicular sperm aspiration) — indications (azoospermia, normal testes, palpable vasa, normal FSH); diagnostic + therapeutic; open = gold standard; never formalin; percutaneous 14 G / 23 G*

Not in the paper answer:

- Biopsy is usually bilateral and planned as testicular sperm extraction (TESE), with an andrology laboratory on site for wet examination and freezing.

ALGORITHM

1. Give a cord block; choose the larger testis (opposite a varicocele).
2. Make a 2 cm incision, open the tunica vaginalis, place stay sutures.
3. Incise the albuginea 3–4 mm, using bipolar on crossing vessels.
4. Take a pea-sized sample with iris scissors (no forceps).
5. Do a touch prep, then a wet prep. If negative, then biopsy the other side.
6. Close the tunica vaginalis with continuous 5-0; no antibiotics.

- A non-absorbable suture marks a positive site for future TESE.

Complication	Point
Haematoma — most common	Prevent with magnification, bipolar and continuous tunica vaginalis closure
Epididymal biopsy — most serious	If that tubule held sperm, obstruction is now certain

4 · VASOGRAPHY

- **Absolute indications (all three)**: azoospermia, complete spermatogenesis on biopsy and at least one palpable vas. Relative: severe oligospermia with normal biopsy, high sperm-bound antibodies, and low volume with poor motility (partial ejaculatory duct obstruction (EDO)).
- Do it **only at planned reconstruction**, because of scar and stricture. It cannot diagnose epididymal obstruction, the commonest non-vasectomy site.

ALGORITHM

1. High scrotal incision; find the vas at the straight–convoluted junction; hemivasotomy at 25×.
2. **Examine vasal fluid first.** If motile sperm, then barbotage and cryopreserve **before any dye**.
3. Pass a 24-G sheath into the abdominal end and inject 1 mL saline. If flow is easy, then it is patent.
4. Rarely, give dilute indigo carmine; blue urine confirms.
5. Use formal contrast only for obstruction above the internal ring.
6. Close with 10-0 mucosa and 8-0 muscularis.

Vasal fluid / vasogram	Meaning and action
No sperm, abdominal end patent	Epididymal obstruction: vasoepididymostomy
Copious fluid, many sperm	Vasal obstruction or EDO: locate it (vasogram, or 2-0 Prolene threaded down)
Block at ejaculatory ducts; both vasa fill from one side	EDO or midline cyst: transurethral resection of the ejaculatory ducts (TURED)
Vas ends blindly	Partial vasal agenesis: cystic fibrosis gene (CFTR) testing; retrieve and freeze
Inguinal block	Inguinal or crossed vasovasostomy
Many sperm, normal vasogram	Retrograde ejaculation, emission failure, aperistalsis

- Use **indigo carmine, not methylene blue**, which kills sperm (use it only after banking).
- Complications are **stricture** (repeated punctures, poor closure, non-water-soluble contrast), devascularisation, haematoma and granuloma.
- **Transrectal seminal vesiculography** aspirates dilated vesicles under transrectal ultrasound (TRUS); sperm there show a patent vas and epididymis, allowing TURED without scrotal exploration.

5 · VASOVASOSTOMY (VASECTOMY REVERSAL)

Evaluation

- Prior fertility proves spermatogenesis. Check follicle-stimulating hormone (FSH) if testes are small or soft; raised FSH means poorer prognosis.
- An indurated epididymis predicts vasoepididymostomy. A **sperm granuloma at the testicular end gives a better prognosis**, because it vents pressure, whatever the interval.
- Operate only if able to do vasoepididymostomy.

Approach and preparation

- The approach is scrotal for most and inguinal for hernia or orchidopexy obstruction. **Never strip the perivasal sheath**.
- **Length**, in escalating steps: dissect to the external then internal ring; free the convoluted vas (4–6 cm; **do not unravel**); reroute beneath the inguinal floor; take the epididymis off the testis. This bridges gaps up to **10 cm**.
- A healthy cut end has a white mucosal ring, soft muscularis and a bleeding edge. **If the muscularis is gritty or bleeding poor, recut.**

- At the testicular end **aspirate, never flush**. Flush saline into the abdominal end to prove patency.

Intraoperative fluid grading — the decision

Vasal fluid	Microscopy	Operation
Copious, crystal clear, watery	No sperm	Vasovasostomy
Copious, cloudy, thin	Sperm with tails	Vasovasostomy
Copious, creamy yellow	Many sperm heads	Vasovasostomy
Scant, granuloma present	Barbotage: sperm	Vasovasostomy
Thick white **toothpaste**	No sperm/heads	**Vasoepididymostomy**
Scant white thin, or dry without granuloma	No sperm/heads	**Vasoepididymostomy**
Any	Grape-like clusters of heads	**Vasoepididymostomy**

> **KEY POINT**
> Clear watery fluid still gets vasovasostomy: the epididymis has not blown out, and sperm usually return. Toothpaste fluid means secondary epididymal blow-out; a discrete level (dilated above, collapsed below) confirms it. When in doubt or inexperienced, do vasovasostomy.

Anastomotic principles

Principle	Reason
Accurate mucosa-to-mucosa	Testicular lumen is 2–5× the abdominal one
Leakproof	Sperm are antigenic; fluid lacks clotting factors, so the seal depends on mucosal sutures
Tension-free	Tension gives early sperm, then falling counts, then azoospermia
Good blood supply; healthy mucosa/muscularis; atraumatic	Recut until bleeding and soft; redo if back-walled or torn

- **Multilayer microdot**: six mapped dots guide **six 10-0 mucosal sutures**, then 9-0 deep muscularis, 9-0 adventitia and 8-0 sheath. The two-layer alternative is 10-0 mucosa and 8-0 full thickness.
- Reinforcing vasal sheath sutures (ReVas) relieve tension and raise patency.
- **Crossed vasovasostomy** suits unilateral inguinal obstruction with a contralateral atrophic testis, or one unreconstructable side.
- **Varicocele with reversal**: vasal vessels are compromised, so stage varicocelectomy ≥6 months later.

Post-operative care and outcomes

- Check semen at **6 weeks**, then 6 months if no pregnancy. Poor motility gets a 4-week prednisone course. Azoospermia at 6 months means redo.
- An anastomotic granuloma heralds obstruction. **Late stricture** shows as falling motility, then count. **Bank sperm once motile sperm appear.**
- With sperm in vasal fluid on at least one side, sperm return in virtually all. Pregnancy is about half, depending on vasectomy interval and female age.

6 · VASOEPIDIDYMOSTOMY

- The epididymis is a single tubule ~200 µm wide and 12–15 feet long, with its thinnest wall at the caput. So a **more distal anastomosis is easier and gives better pregnancy**.
- **Indications**: reversal by the fluid criteria; or non-vasectomy OA with proven spermatogenesis, no vasal sperm and a patent vas and ejaculatory ducts.

ALGORITHM

1. Hemitransect the vas at the straight–convoluted junction; sample fluid; confirm patency.
2. Open the tunica vaginalis; choose a site **above** the obstruction where tubules are dilated. If unclear, then puncture distal to proximal until sperm appear.
3. Make a 3–4 mm tunic window.
4. Bring the vas through the tunica vaginalis, **fixed with 5-0 Prolene to the albuginea (tension relief)**.
5. Place posterior tunic–vas 8-0, then intussusception sutures, then outer tunic–vas adventitia.

Technique	For	Against
Longitudinal intussusception (LIVE) — used for all	Two 10-0 longitudinal sutures placed in the **still-distended** tubule; incise between; tubule pulled into vas; bloodless, leakproof	Fluid not checked before sutures set
End-to-side	Bloodless; fluid checked first	Suturing a collapsed tubule
End-to-end	Rapid level finding; bridges gap	Haemostasis hard; inferior epididymal supply lost

- **No simultaneous varicocelectomy**; stage it after 6 months.
- Late shutdown occurs, so **bank sperm during surgery and once they appear**. If it fails, redo or retrieve sperm for ICSI.

7 · TURED

P3 *Ejaculatory duct obstruction — congenital (midline cyst, atresia) vs acquired; low volume, acidic, low fructose, palpable vas, normal FSH; TRUS; TURED only after couple declines IVF; sperm return in ~half; cold-knife reobstructs; reflux, epididymitis, retrograde ejaculation; vesiculoscopy*

Not in the paper answer:

- For retrograde ejaculation give pseudoephedrine 60 mg 90 min before, or use alkalinised urinary sperm.

ALGORITHM

1. Use a 24-Fr resectoscope with a finger in the rectum.
2. Resect the **verumontanum** to open the dilated orifices or cyst, sparing bladder neck, sphincter and rectum.
3. **Blue efflux confirms** adequacy; coagulate minimally.
4. Catheter overnight; 7 days of antibiotics.

8 · SPERM RETRIEVAL

P2 *Sperm retrieval techniques (sperm extraction in obstructive azoospermia; testicular aspiration and assisted reproduction) — ICSI essential; PESA/MESA/TESA/TESE/micro-TESE table; MESA by capillary action; micro-TESE ~half of NOA; retrieval by diagnosis (AZFa/b **0% — do not operate**); frozen = fresh except SCI*

Not in the paper answer:

- Microsurgical epididymal sperm aspiration (MESA) gives most motile sperm and multiple vials. **Microdissection TESE (micro-TESE) is best in NOA**.

KEY POINT

Paradox: in obstruction the **most motile sperm are proximal** (efferent ductules); distal tubules hold senescent fragments and macrophages. Yet fertilisation is best when the longest epididymis is preserved.

ALGORITHM

1. Micro-TESE: general or regional anaesthesia.
2. At 10×, find the avascular plane on the anterior albuginea; make a wide equatorial incision and bivalve the testis.
3. Excise larger tubules; an andrologist in theatre examines them; map positives.
4. If negative, then explore the other testis.
5. Close the albuginea with 5-0 and the tunica vaginalis with 4-0.

9 · CRYOPRESERVATION

- Freeze at diagnostic biopsy, at every reconstruction (before any dye), and from the ejaculate once motile sperm return after reversal.

10 · VARICOCELECTOMY

P2 P3 *Surgical management of bilateral varicocele; Varicocele: controversies — indications; approach table (microsurgical recurrence <1%, hydrocele ~0); micro-Doppler, testis delivery, impulse test; complications*

Not in the paper answer:

- Conventional inguinal repair has hydrocele 3–15% and recurrence 5–15%, because it misses scrotal and periarterial veins.
- Ligate vasal veins only if >2.5 mm, keeping one set: **one intact deferential set guarantees venous return**.

ALGORITHM

1. Local anaesthesia with sedation; 2–3 cm incision over the external ring.
2. Deliver the cord on a Penrose, excluding the ilioinguinal nerve and genital branch.
3. **Map arteries with micro-Doppler before handling.**
4. Open the spermatic fasciae; work at 10–25× with 1% papaverine; loop the artery.
5. Ligate **all veins except vasal**; preserve lymphatics.
6. Deliver the testis; ligate external spermatic and gubernacular veins.
7. Do the impulse test. If there is no impulse, then a vein was missed.

- **End point**: only testicular and cremasteric arteries, lymphatics, and the vas with its vessels remain.

KEY POINT

Recurrence means a vein was missed; hydrocele, a lymphatic tied; atrophy, an artery tied. Microsurgery with testicular delivery addresses all three.

11 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Madgar (1995)	Randomised controlled trial (RCT): varicocelectomy vs no surgery in infertile men	1-year pregnancy 44% vs 10%	Repair of clinical varicocele with abnormal semen improves fertility

71 Physiology of Penile Erection and Pathophysiology of Erectile Dysfunction

1 · DEFINITION AND KEY CONCEPTS

- **Erectile dysfunction (ED)** is the consistent or recurrent inability to attain and/or maintain an erection sufficient for sexual satisfaction (American Urological Association (AUA) 2018).
- Erection needs **sinusoidal (trabecular) smooth muscle relaxation, arterial dilation and venous compression (veno-occlusion)**. Rigidity comes from trapping blood, not from inflow alone.
- Milestones MCQ: nitric oxide (NO) was identified in the 1990s, and oral phosphodiesterase type 5 inhibitors (PDE5Is) arrived in 1998.

2 · FUNCTIONAL ANATOMY

Corpora

- There are paired dorsal **corpora cavernosa** and a ventral **corpus spongiosum**, which surrounds the urethra and expands as the glans.
- The septum is **incomplete** in humans, so one injection fills both corpora.
- Sinusoidal oxygen tension (Po2) is venous when flaccid (**~35 mm Hg**) and arterial when erect (**~90 mm Hg**).

Tunica albuginea

- The **inner circular layer** gives off **intracavernous pillars** (absent at 5–7 o'clock). The **outer longitudinal layer** is thicker dorsally and absent in the ventral groove.
- The ventral tunica is ~0.8 mm against ~2.2 mm dorsally. So 5–7 o'clock is the weak zone for **penile fracture** and implant crossover or perforation.
- **Veno-occlusion**: emissary veins run obliquely between the layers and are compressed as the sinusoids stretch the tunica. Cavernous arteries escape because they lie in a periarterial sheath.
- The spongiosal tunica is thin, single-layered and absent in the glans, so spongiosum and glans are **low-pressure arteriovenous shunts** and never become rigid.

Arteries

- The internal iliac gives the **internal pudendal**, which becomes the common penile artery with three branches.
- The **cavernous** artery is the main corporal supply. Its **helicine arteries** are tortuous when flaccid and straight when erect, and open into sinusoids without capillaries.
- The **dorsal** artery fills the glans, and the **bulbourethral** artery supplies the bulb and spongiosum.
- **Accessory pudendal arteries** occur in ~one-third and may be dominant. Preserving them at radical prostatectomy speeds erectile recovery.

Veins

Sinusoids drain into the **subtunical venous plexus**, which exits through **emissary veins** in the tunica.

System	Drains	Destination
Deep dorsal vein (with or without circumflex and periurethral veins)	Glans and distal two-thirds of corpora	Periprostatic (Santorini) plexus
Cavernous and crural veins	Proximal (infrapubic) penis	Internal pudendal veins
Superficial dorsal vein	Skin, subcutaneous tissue	Saphenous veins

> **KEY POINT**
>
> Vascular anatomy varies widely between men, so the same "nerve-sparing" operation gives different erectile outcomes.

3 · HAEMODYNAMICS: PHASES OF ERECTION AND DETUMESCENCE

P1 *Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED — sequence of events and phase table (flaccid, latent, tumescence, full, rigid, detumescence)*

Not in the paper answer:

- Full-erection intracavernous pressure (ICP) plateaus at **~100 mm Hg**; the rigid phase is suprasystolic, driven by the pudendal nerve via ischiocavernosus.
- **Trap**: arterial inflow peaks in the latent phase, not at rigidity. This is why peak systolic velocity (PSV) on Doppler is measured early.
- Detumescence has three phases: initial (transient ICP rise), slow (veins gradually reopen) and fast (outflow restored).

4 · NEUROANATOMY

Peripheral and spinal

	Sympathetic	Parasympathetic	Somatic
Origin	**T11–L2** (often T10–T12)	**S2–S4** intermediolateral columns	S2–S4; motor from **Onuf nucleus**
Route	Sympathetic chain, superior hypogastric plexus, hypogastric nerves, then pelvic plexus	Pelvic nerves (nervi erigentes) to the pelvic plexus	Pudendal nerve; afferent via dorsal nerve of penis
Role	**Detumescence**; emission	**Erection** (cavernous nerves)	Sensation; ischiocavernosus (rigidity), bulbocavernosus (ejaculation)

- **Cavernous nerves** arise from the pelvic plexus, carry both autonomic types and run posterolateral to the prostate. Pelvic ganglia lie along the bladder–prostate junction and seminal vesicles, so they are vulnerable in radical pelvic surgery.
- The **bulbocavernosus reflex** (glans squeeze contracts bulbocavernosus) has limited value in ED work-up.

Three types of erection

Type	Trigger	Mechanism	Spinal cord injury (SCI)
Psychogenic	Audiovisual, fantasy	Brain modulates spinal centres, chiefly by suppressing sympathetic tone	May persist with sacral lesions; absent with lesions above T9
Reflexogenic	Genital touch	Sacral reflex arc to the cavernous nerves	Preserved in upper cord lesions
Nocturnal	Rapid eye movement (REM) sleep	Cholinergic pontine tegmentum on; locus coeruleus (noradrenergic) and raphe (serotonergic) silent	—

Supraspinal (central)

- The **medial preoptic area (MPOA)** is dopaminergic and integrates hormonal and sensory cues. The **paraventricular nucleus (PVN)** sends pro-erectile **oxytocin** neurons to the lumbosacral nuclei, and the nucleus paragigantocellularis is the **serotonergic brake**.

Central neurotransmitters

P1 *Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED — promoters (dopamine, oxytocin, NO, noradrenaline, melanocortins, 5-HT2C) vs inhibitors (5-HT1A/2, GABA, opioids, cannabinoids, prolactin)*

Not in the paper answer:

- Apomorphine is a D1/D2 agonist acting in the PVN, which releases oxytocin and then NO. Melanocortins act on the MC4 receptor.
- Serotonin 5-HT1A and 5-HT2 receptors inhibit erection but facilitate ejaculation.

> **KEY POINT**
> NO synthase (NOS) inhibitors block both apomorphine- and oxytocin-induced erections, so NO is the final common central pathway. **Trap**: noradrenaline is pro-sexual centrally but the principal anti-erectile transmitter peripherally.

5 · MOLECULAR MECHANISMS OF SMOOTH MUSCLE TONE

Every erectogenic drug acts on one of these pathways.

Contraction (flaccid state)

- Corporal smooth muscle is **contracted at rest** through myogenic tone, adrenergic tone and endothelial constrictors.
- **Calcium pathway**: noradrenaline (α1), endothelin-1 (ETA), angiotensin II (AT1), PGF2α and thromboxane A2 act via Gq and phospholipase C to form inositol trisphosphate (IP3), raising Ca^{2+}. Ca^{2+}–calmodulin then activates **myosin light chain (MLC) kinase**, which phosphorylates MLC20.
- **Ca^{2+} sensitisation by RhoA/Rho-kinase**: it inhibits **MLC phosphatase (MLCP)**, giving sustained contraction without a further Ca^{2+} rise.
- Rho-kinase is raised in diabetes mellitus (DM), hypertension, hypercholesterolaemia, hypogonadism, ageing and nerve injury, and it also drives fibrosis.

Relaxation (erection)

- **NO–cGMP**: NOS converts L-arginine and O2 into NO and citrulline. **Neuronal NOS initiates and endothelial NOS (eNOS) sustains** erection.
- NO activates soluble guanylyl cyclase to make cyclic GMP (cGMP), which activates **protein kinase G (PKG-I)**. PKG lowers intracellular Ca^{2+}, inhibits Rho-kinase and activates MLCP, so the muscle relaxes. PKG-I knockout mice cannot erect, so PKG is the bottleneck.
- **cAMP pathway**: PGE1, vasoactive intestinal peptide (VIP) and calcitonin gene-related peptide (CGRP) act via Gs to raise cyclic AMP (cAMP), which activates protein kinase A (PKA). It is **NO-independent**, which is why intracavernosal alprostadil works when nerves or endothelium fail.
- **PDE5 (cGMP-specific) is the principal cavernous PDE**. PDE6 is retinal (visual side effects); PDE2, 3 and 4 degrade cAMP. Papaverine is nonspecific.
- **Gap junctions (connexin-43)** synchronise relaxation, so a local input gives a uniform erection.
- Oxygen acts as a switch: erect Po2 >50–60 mm Hg permits full NOS activity, while low flaccid Po2 suppresses it.
- Excess **adenosine** with increased A2B signalling causes prolonged erection in adenosine deaminase-deficient and sickle cell mice, the **priapism models**.
- Acetylcholine acts indirectly (endothelial NO, presynaptic adrenergic inhibition), so **atropine does not abolish** neural erection MCQ.

Second messenger	Activators	Kinase	Degraded by	Drug
cGMP	NO, CO, H2S, natriuretic peptides	PKG-I	PDE5	PDE5Is
cAMP	PGE1, VIP, CGRP, adenosine (A2)	PKA	PDE2, 3, 4	Alprostadil; papaverine (both)

Detumescence

NO release stops and PDE5 hydrolyses cGMP. Sympathetic noradrenaline (α1; the principal transmitter of flaccidity), endothelin-1 and Rho-kinase then contract the trabeculae, so the veins

reopen.

> **KEY POINT**
>
> Corporal tone is a tug-of-war over one phosphorylation site (MLC20): Ca^{2+}–calmodulin–MLC kinase and Rho-kinase keep it phosphorylated; NO–cGMP–PKG removes the phosphate. **Trap**: "MLCP" means the phosphatase.

6 · CLASSIFICATION OF ED

Approach	Categories
Neurovascular mechanism	Failure to **initiate** (neurogenic, psychogenic, endocrine) · to **fill** (arterial) · to **store** (veno-occlusive)
Aetiological	Psychogenic · neurogenic · hormonal · arteriogenic · cavernosal (venogenic) · drug-induced · systemic disease · mixed (commonest)

Most organic ED has a psychogenic component.

7 · RISK FACTORS AND ENDOTHELIAL DYSFUNCTION

ED often signals vascular disease before the heart does.

- **Age** is the strongest demographic predictor, but DM, heart disease and hypertension carry more risk (Massachusetts Male Aging Study (MMAS)).
- **Endothelial dysfunction**: oxidative stress causes **NOS uncoupling**, while arginase and asymmetric dimethylarginine (ADMA) further lower NO.
- **Artery-size hypothesis**: smaller penile arteries become symptomatic first. ED preceded coronary artery disease (CAD) by a mean of **~2 years**, so ED is a **sentinel** and needs cardiac risk stratification (see Ch 72).
- The common end-path is raised Rho-kinase and transforming growth factor-β (TGF-β), which lower the smooth muscle-to-collagen ratio and cause veno-occlusive failure.

8 · PATHOPHYSIOLOGY BY TYPE

Psychogenic

- The brain inhibits the spinal centres, and excess sympathetic tone contracts the muscle. Causes include performance anxiety, partner conflict, depression and post-traumatic stress disorder (PTSD).

Neurogenic

- **Brain** causes include Parkinson disease, stroke, tumours and dementia. **Cord** causes include SCI (commonest), spina bifida, disc disease and multiple sclerosis (MS).

Complete lesion	Reflexogenic erection	Psychogenic erection
Upper cord (above sacral centre)	Preserved in **~95%**; brief, need continuous stimulation	Lost
Lower cord (sacral)	Preserved in only **~25%**	May occur (thoracolumbar pathway); marginal tumescence

- **Iatrogenic**: radical prostatectomy, cystoprostatectomy, abdominoperineal resection (APR) and pelvic radiation. Most men have at least a transient decline despite nerve sparing, worse with baseline ED and age >60.
- Cavernous nerve injury causes apoptosis and fibrosis, leading to venous leak. This is the rationale for early penile rehabilitation.
- **Pelvic fracture** injures the cavernous nerves or pudendal artery. With posterior urethral injury, early realignment preserves potency better than delayed anastomosis.

Hormonal

- **Testosterone (T)** governs **libido** more than mechanics, but it maintains NOS, PDE5 and smooth muscle. Low libido is the commonest sexual symptom of low T.
- AUA: measure **morning, fasting** T in all men with ED. Low T is **<300 ng/dL**; treat only with ≥2 low levels and symptoms. Nocturnal erections need ~200 ng/dL.
- Nocturnal erections are androgen-dependent, but erections to visual stimuli are largely independent MCQ. Dihydrotestosterone (DHT) is the key penile androgen.
- **Hyperprolactinaemia** (prolactinoma, antipsychotics) suppresses gonadotrophin-releasing hormone (GnRH), causing low T and low libido.
- A low T:oestradiol ratio (obesity), hyperthyroidism and hypothyroidism also cause ED.

Vasculogenic – arteriogenic

- Atherosclerotic or traumatic occlusion of the hypogastric–pudendal–cavernous tree lowers perfusion pressure, giving a slower, softer erection.
- Risks are hypertension, hyperlipidaemia, smoking, DM, obesity, pelvic irradiation and blunt perineal trauma. Trauma causes focal stenosis in young men, who are revascularisation candidates.
- Chronic hypoxia raises TGF-β1 and collagen, so "inflow failure becomes storage failure".
- In hypertension, angiotensin receptor blockers (ARBs) and nebivolol protect penile tissue.

Vasculogenic – veno-occlusive (venous leak)

- Causes are **tunical** (ageing, DM, Peyronie disease, penile fracture), **fibroelastic trabecular change** preventing sinusoidal expansion, **insufficient relaxation** (anxiety, adrenergic tone, endothelial dysfunction), and **acquired shunts** after priapism shunt surgery. Congenital ectopic or crural veins cause primary ED.

> **KEY POINT**
> Veno-occlusion is an emergent property of healthy tissue (elastic tunica and fully relaxed trabeculae). So most "venous leak" is really cavernosal smooth muscle disease, which is why venous ligation surgery fails and is investigational.

Metabolic and systemic disease

- **DM** causes the earliest, most severe ED through arteriopathy, nitrergic neuropathy, endothelial dysfunction, advanced glycation end products (AGEs), raised Rho-kinase and hypogonadism. **Key**: the NO source is damaged but the response to exogenous NO is preserved, so PDE5Is work until fibrosis sets in.
- **Metabolic syndrome and obesity** improve with weight loss, Mediterranean diet, exercise and bariatric surgery. In **hyperlipidaemia**, oxidised low-density lipoprotein (LDL) inhibits NO, and statins are linked to better International Index of Erectile Function (IIEF) scores.
- **Chronic kidney disease**: up to **80%** of men with end-stage renal disease have ED, and it persists in ~half after transplant.
- Other associations include HIV, cirrhosis, chronic obstructive pulmonary disease, obstructive sleep apnoea and lower urinary tract symptoms.

Drug-induced

Class	Cause ED	Mechanism / note	Alternative
Antihypertensives	**Thiazides**, **nonselective β-blockers**, clonidine	Thiazide – unknown; β-blocker – prejunctional α2	ACE inhibitors, ARBs, nebivolol, α1-blockers (ejaculatory dysfunction), calcium channel blockers

Class	Cause ED	Mechanism / note	Alternative
Antidepressants	Selective serotonin reuptake inhibitors (SSRIs; paroxetine worst), tricyclics	Spinal 5-HT2/5-HT3 inhibition, less dopamine and NO; post-SSRI dysfunction may persist	Bupropion, mirtazapine, nefazodone, buspirone; add a PDE5I
Antipsychotics	Typical > atypical	Dopamine block causes hyperprolactinaemia	Newer agents
Antiandrogens	LHRH agonists/antagonists, androgen receptor antagonists, cyproterone, **5α-reductase inhibitors (5ARIs)**	5ARI: least T effect yet persistent ED after stopping	None
Recreational	Tobacco (dose–response, ≥20 pack-years), heavy alcohol, opioids	Endothelial damage, neuropathy, hypogonadism	Stop or moderate
Others	Cimetidine, spironolactone, carvedilol	—	—

Trazodone causes priapism, and among anticonvulsants carbamazepine and phenytoin are worse than lamotrigine.

Primary (lifelong) ED

- It is usually psychogenic, but exclude hypoplastic cavernous arteries, congenital venous leak, corporal atresia and micropenis.

> **KEY POINT**
> Why ED is a sentinel: nearly every systemic disease reaches the penis through the same three doors — endothelial NO failure, Rho-kinase-driven contraction and TGF-β-driven fibrosis.

72 Evaluation and Management of Erectile Dysfunction

1 · DEFINITION AND SIGNIFICANCE

- **Erectile dysfunction (ED)** (American Urological Association, AUA): a consistent or recurrent inability to attain and/or maintain an erection sufficient for sexual satisfaction. Minimum **3 months**, except after trauma or surgery.
- About **20%** of adult men have ED (up to 15% at 40–49 y; up to 40% at 60–69 y).
- **ED predicts cardiovascular disease (CVD)** through shared endothelial dysfunction, as strongly as smoking or a family history of myocardial infarction (MI). Younger onset means greater cardiovascular (CV) risk.

Independent association (multivariate)	Risk factors
Strongest	Antidepressant use, then antihypertensive use
Strong	Diabetes mellitus (DM), the strongest disease; obstructive urinary symptoms
Moderate	Hypertension, prostate enlargement, smoking, body mass index (BMI) ≥30, inactivity
Weak/null after adjustment	CVD, hypercholesterolaemia

- Drugs cause up to 25% of ED (see Ch 71); opioids double the risk.

2 · MANAGEMENT PRINCIPLES

- **Shared decision making** (AUA 2018): all non-contraindicated options are offered **in parallel, not ranked**, and the patient may choose any first. They are phosphodiesterase type 5 inhibitors (PDE5Is), the vacuum erection device (VED), intraurethral (IU) alprostadil, intracavernosal injection (ICI) and prosthesis.
- In testosterone (T) deficiency (symptoms and total T below 300 ng/dL), restoring T probably improves the other ED treatments.

Cardiac risk assessment – Princeton Consensus

P1 *Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED — Princeton high (defer sex and ED treatment) / indeterminate (stress test, restratify) / low (treat)*

Not in the paper answer:

- Every man with ED is at potential CV risk; sex equals moderate exertion. **Refer** after failed therapy, for a young trauma patient, for deformity, in medicolegal cases, and for complex endocrine, psychiatric or vascular problems.

3 · BASIC EVALUATION

History

- Ask about onset, pattern, morning erections, libido, ejaculation, curvature and prior treatment.
- Ask about DM, CVD, neurological disease, pelvic surgery or radiation, trauma and drugs. Screen for depression.

Psychogenic	Organic
Sudden onset	Gradual onset
Complete, immediate loss	Incremental progression
Situational	Global
Waking/nocturnal erections present	Poor or absent

> **KEY POINT**
> Preserved nocturnal/morning erections prove an intact neurovascular axis, the strongest historical pointer to psychogenic ED.

Questionnaires

- The **International Index of Erectile Function (IIEF)** has 15 items in 5 domains: erectile function, orgasm, desire, intercourse satisfaction and overall satisfaction.

P1 *Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED — IIEF-5/SHIM bands: 5–7 severe … 22–25 no ED*

- The Sexual Health Inventory for Men (SHIM) has 5 items scored 1–5 over the past 6 months. Questionnaires grade severity and response but do **not** identify the cause.

Physical examination

- Look for hypogonadism (testes, gynaecomastia), vascular disease (blood pressure (BP), obesity, pulses), neuropathy (genital sensation, bulbocavernosus reflex) and structural problems (plaque, micropenis, chordee).

Laboratory tests

- **Recommended**: fasting glucose or glycated haemoglobin (HbA1c), lipids, serum chemistry, full blood count and **morning (7–11 am) total testosterone**.
- Repeat a low T, because about one-third of mildly low values normalise, and add luteinising hormone (LH) and prolactin. Check prostate-specific antigen (PSA) if prostate disease is suspected or before T therapy.

KEY POINT

The basic work-up finds the disease behind the ED (undiagnosed DM, dyslipidaemia, hypogonadism, CV risk); it does not prove ED.

4 · HORMONAL EVALUATION

P4 *Male menopause (andropause, late-onset hypogonadism) — bioavailable T = free + albumin-bound; SHBG rises with age; LH separates primary/secondary; prolactin if LH low; pituitary MRI if T <150 ng/dL*

Not in the paper answer:

- The AUA threshold is **300 ng/dL (10.4 nmol/L)**; above about 230 ng/dL, sexual function correlates only weakly with T. Order a pituitary magnetic resonance imaging (MRI) scan if T is **<150 ng/dL**.
- Hyperprolactinaemia is caused by antipsychotics, tricyclics, opiates, prolactinoma, hypothyroidism and renal failure.

5 · SPECIALISED TESTING

These tests are for young trauma patients, primary ED, surgical candidates, medicolegal cases and failed therapy.

Test	Line	Answers	Indication
Combined intracavernosal injection and stimulation (CIS)	1st	Can the vessels produce a rigid erection?	Screening
Penile duplex Doppler	2nd	Arterial or veno-occlusive? Asymmetry?	Most reliable, least invasive
DICC (dynamic infusion cavernosometry/cavernosography)	3rd	Is there a venous leak, and where?	Trauma, lifelong ED, before venous surgery
Selective internal pudendal arteriography	3rd	Site of arterial lesion; is the inferior epigastric donor usable?	Young trauma patient before revascularisation

CIS test

- Inject alprostadil 10–20 µg or Bimix/Trimix 0.3 mL into the lateral base, then compress, stimulate and rate rigidity. A rigid, sustained erection means normal haemodynamics; anxiety or a low dose gives false negatives, so redose.
- If there is no detumescence by about 1 h, inject diluted phenylephrine (500 µg/mL) every 3–5 min.

Penile duplex Doppler

- Measure cavernous artery peak systolic velocity (PSV) and end-diastolic velocity (EDV) at the base every 5 min up to 20 min after ICI.

Parameter	Arterial insufficiency	Veno-occlusive dysfunction
PSV	**<25 cm/s** (25–35 borderline; >35 normal)	Stays >25 cm/s
EDV	—	**>5 cm/s**
Resistive index (RI) = (PSV − EDV)/PSV	—	**<0.75** (normal >0.9)
Other	Acceleration time >122 ms; side-to-side PSV difference >10 cm/s; artery dilation <75%	Rapid detumescence
When assessed	Within 5 min	15–20 min (needs full relaxation)

KEY POINT

Venous leak can be diagnosed only with adequate inflow and full smooth muscle relaxation; an anxious patient mimics veno-occlusive dysfunction, so redose first. Anatomical variants can falsely suggest arteriogenic ED.

DICC, NPTR and neurological tests

- DICC measures flow-to-maintain, pressure decay and cavernous artery systolic occlusion pressure after a vasodilator; cavernosography localises the leak.
- Nocturnal penile tumescence and rigidity (NPTR) testing in a sleep lab is the standard. The home **RigiScan** measures **radial** rigidity; normal is 4–5 erections a night, ≥60% rigid for ≥10 min. It suits obscure cause, medicolegal cases and suspected psychogenic ED.

6 · TREATMENT

ALGORITHM

1. Treat reversible causes (lifestyle, drugs, hormones, psychosocial) and stratify CV risk.
2. Shared decision: PDE5I, VED, IU alprostadil or ICI. Low-intensity extracorporeal shock wave therapy (Li-ESWT) is investigational.
3. If the response is inadequate, then optimise, combine or switch.
4. If still inadequate, then offer a penile prosthesis (see Ch 75).

Lifestyle and risk-factor modification

- Stopping smoking, exercise, weight loss and a Mediterranean diet improve function; control DM, lipids and BP.
- **Change medication**: replace thiazides or β-blockers with calcium channel blockers, angiotensin-converting enzyme inhibitors or angiotensin receptor blockers. For a selective serotonin reuptake inhibitor (SSRI), switch to bupropion or mirtazapine, or add a PDE5I.
- **Psychosexual therapy** works best combined with medical therapy.

Hormonal therapy

- Give T only for confirmed deficiency, as a ~3-month trial; it improves libido more than erection and is synergistic with PDE5Is.
- Monitor digital rectal examination, PSA, haematocrit, liver tests and lipids at baseline, 3–6 months and yearly. The main adverse effect (AE) is erythrocytosis; randomised trials show no causal CV link.
- Preparations include gels (transfer risk), intramuscular enanthate/cypionate (200–250 mg every 2–3 weeks) and undecanoate depot (750 mg at 0 and 4 weeks, then every 10 weeks). Avoid methyltestosterone because it is hepatotoxic.
- T does not correct hyperprolactinaemia: stop the drug, give bromocriptine for prolactinoma, and operate for failure or visual compromise.

PDE5 inhibitors (first-line oral)

They block cyclic guanosine monophosphate (cGMP) breakdown, so they **enhance, not initiate**, erection; sexual stimulation is obligatory.

	Sildenafil	Vardenafil	Tadalafil	Avanafil
Doses (mg)	25, 50, 100	5, 10, 20	5, 10, 20 (2.5/5 daily)	50, 100, 200
Time to peak, T (h) max	0.8 (~1 h before)	0.7–0.9 (~1 h)	**2** (~2 h)	**0.3–0.5** (~30 min)
Onset (min)	15–60	15–60	15–120	15–60
Half-life (h)	3–5	4–5	**17.5**	3–5

	Sildenafil	Vardenafil	Tadalafil	Avanafil
Fatty food	Lowers absorption	Lowers absorption	**No effect**	Lowers absorption
Characteristic AE	Visual (blue tinge; PDE6), flushing	Visual, flushing; **QT prolongation, so avoid with class 1A/3 antiarrhythmics** and congenital long QT	**Back pain/myalgia** (PDE11)	Fewest visual effects
Nitrates	**Contraindicated (all)**			

- Headache is the commonest AE. Efficacy is equal across agents: about 70% achieve intercourse, and 40–50% in DM and after prostatectomy.
- **Nitrates** in any form, including amyl nitrite "poppers", cause profound hypotension. If angina occurs, give no nitrates for **24 h** (sildenafil, vardenafil) or **48 h** (tadalafil). There is no antidote; treat hypotension with head-down tilt, fluids and phenylephrine.
- Use caution with α-blockers (stabilise first, start low) and aortic stenosis; CYP3A4 inhibitors need a lower dose.
- They are not studied after MI or stroke within 6 months, with BP <90/50 or >170/100, or in retinitis pigmentosa. For sudden visual loss, stop the drug and seek care.
- **Before labelling failure**, check stimulation, timing, food, maximum dose and **up to 9–10 attempts**; correct hypogonadism, glycaemia and lipids; then switch agent or try daily tadalafil.
- **Penile rehabilitation** after prostatectomy is unproven, since REINVENT and other nightly-dosing trials were negative.

Other oral agents (yohimbine, apomorphine, trazodone) have little or no proven benefit.

Vacuum erection device

- Negative pressure fills the corpora and a base ring traps the blood; **remove it within 30 min**. The erection is hinged at the base, cool and cyanotic.
- Up to 90% get an erection but only 30–70% are satisfied; it suits older men in stable relationships. Take care with anticoagulants.

Intraurethral alprostadil (MUSE)

- A prostaglandin E1 (PGE1) pellet of 125–1000 µg is inserted into the urethra after voiding. About 50% respond, so it is a niche option: PDE5I failure after nerve injury, add-on to a PDE5I, or soft glans after prosthesis.
- Urethral pain affects about one-third; use a condom if the partner is pregnant.

Intracavernosal injection

ICI acts directly on smooth muscle, **independent of nerves and nitric oxide (NO)**, so it works when PDE5Is fail (neurogenic, post-prostatectomy, DM).

Agent	Mechanism	Notes
Alprostadil (PGE1) 5–40 µg	Adenylate cyclase raises cyclic AMP (cAMP)	Only FDA-approved ICI; ~70% efficacy; metabolised locally; **penile pain** commonest AE; least priapism and fibrosis
Papaverine	Nonspecific PDE inhibitor; Ca^{2+} channel block	Cheap; raises liver enzymes, causes priapism and fibrosis, so not used alone
Phentolamine	α-antagonist	Adjunct only; hypotension, reflex tachycardia
Trimix (alprostadil 10 µg, papaverine 30 mg and phentolamine 1 mg per mL)	Synergistic	≈90%; rescues monotherapy failures; less pain

Agent	Mechanism	Notes
Bimix	Papaverine with phentolamine, or alprostadil with phentolamine	≈90%

- Titrate in the office from a low dose, since neurogenic ED is very sensitive, and inject the lateral shaft.
- **Priapism** is highest with papaverine mixes and in neurogenic or young men; the patient must seek care beyond 4 h. Long-term use causes fibrosis and curvature.
- Contraindications: psychological instability, priapism risk (sickle cell), severe coagulopathy, unstable CVD, poor dexterity (the partner may inject), and monoamine oxidase (MAO) inhibitors, because phenylephrine reversal risks a hypertensive crisis.

Topical agents (alprostadil cream, the over-the-counter MD3000 gel) give modest benefit.

Restorative and investigational

Therapy	Proposed mechanism	Evidence/status
Li-ESWT	New vessels; nerve, endothelial, smooth muscle regeneration	RCT meta-analysis: IIEF better than sham in **mild–moderate vasculogenic** ED, the best evidence; still investigational (Sexual Medicine Society of North America: research only)
Platelet-rich plasma	Growth factors	Conflicting RCTs
Stem cells	Regeneration	Uncontrolled series only
Botulinum toxin ICI	Blocks sympathetic noradrenaline	Short-term benefit in refractory ED

Combination therapy

Monotherapy fails in nearly 40%. Useful pairs are daily tadalafil with on-demand sildenafil, a PDE5I with T, IU, ICI or VED, and a prosthesis with IU for a soft glans. PDE5I with ICI is off-label, so test it in the office.

Surgery

- A **penile prosthesis** suits men whose other therapy is contraindicated, ineffective or unwanted, and those with corporal fibrosis or Peyronie disease (see Ch 75).

P2 *Surgical management of erectile dysfunction and penile prosthesis — arterial revascularisation (age <55, nonsmoker, nondiabetic, focal post-traumatic pudendal stenosis; glans hyperaemia) and investigational venous ligation*

7 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Prostate Cancer Prevention Trial (placebo arm)	Prospective cohort, men ≥55	ED gave 45% more cardiac events at 5 years	ED is a CV sentinel; stratify risk
REINVENT	RCT: nightly vs on-demand vardenafil vs placebo after nerve-sparing prostatectomy	No difference in spontaneous recovery	Scheduled "rehabilitation" PDE5I is unproven

73 Priapism

1 · DEFINITION AND CLASSIFICATION

- **Priapism** is a full or partial erection lasting **≥4 h** beyond, or unrelated to, sexual stimulation. Untreated ischaemic priapism causes fibrosis and permanent erectile dysfunction (ED).
- It is classified by **oxygenation, not duration**: failed outflow causes time-critical ischaemia, while excess arterial inflow gives an oxygenated, painless erection that can wait.

Feature	Ischaemic (low-flow)	Stuttering (recurrent ischaemic)	Non-ischaemic (high-flow)
Share	**95%**	Subset of ischaemic	<5%
Lesion	Failed outflow, like a compartment syndrome	Repeated self-limiting episodes, often nocturnal	Fistula between cavernosal artery and sinusoids
Typical cause	Drugs, sickle cell disease (SCD), blood disorders	SCD	Blunt perineal/penile trauma, needle injury
Pain / rigidity	Painful (within 6–8 h); fully rigid, tender	Painful; rigid during episode	Painless; partially rigid
Blood gas / Doppler	Low oxygen, high CO_2, acidotic; absent or low flow	Ischaemic in each episode	Arterial gas; high flow, fistula or blush
Management	Emergency decompression	Treat episode as ischaemic, plus prophylaxis	Observe, then embolise
Fibrosis/ED	Common if untreated	Risk with each episode	Usually absent

> **KEY POINT**
>
> Trap: stuttering priapism is **ischaemic**, so any episode lasting ≥4 h needs the same first-line treatment.

2 · AETIOLOGY

P3 *Priapism: Diagnosis and Management — causes: ICI, α-blockers, trazodone, antipsychotics, cocaine, anticoagulant withdrawal, SCD, leukaemia, metastases, venom; non-ischaemic trauma with delayed onset*

Not in the paper answer:

- Metastases obstruct venous outflow.

Ischaemic – extra points

- Many other drugs, including oral phosphodiesterase type 5 inhibitors (PDE5Is), can cause it, so take a full drug history.
- In leukaemia, give cytotoxics and leukapheresis alongside penile treatment.
- **Recurrence after a shunt** may mean shunt thrombosis. Give **aspirin 325 mg + heparin 5000 U subcutaneously** before surgery, then aspirin 81 mg daily for 2 weeks.

Sickle cell disease

- Deoxygenated haemoglobin S polymerises, causing haemolysis and low nitric oxide (NO), which obstruct outflow.
- Lifetime risk is **29–42%**; SCD causes 23% of adult and **63% of paediatric** ischaemic priapism. Stuttering is commoner than major episodes.

Non-ischaemic

- Onset is often delayed ≥1 day after trauma, because the clot over the arterial tear disperses in a nocturnal erection.
- Transient post-ischaemic hyperaemia after treatment mimics it; colour Doppler distinguishes them.

3 · MOLECULAR PATHOPHYSIOLOGY

- Low basal NO reduces cyclic guanosine monophosphate (cGMP) feedback, so **PDE5 is down-regulated**. A neural NO surge then produces cGMP that cannot be broken down, so relaxation is unopposed.
- Paradox: **a daily PDE5I restores PDE5 and prevents** recurrent episodes.

4 · EVALUATION

P3 **P1** *Priapism: Diagnosis and Management; Doppler in urology — history, examination, corporal blood gas table (ischaemic pO_2 <30, pCO_2 >60, pH <7.25; flaccid 40/50/7.35; arterial >90/<40/7.4), PDUS (PSV <50 vs >50 cm/s), parallel labs*

Not in the paper answer:

- Record baseline erectile function, because it is needed for prosthesis counselling.
- The glans and spongiosum stay soft in both types; their involvement suggests another disease.

ALGORITHM

1. History, examination and corporal blood gas.
2. If ischaemic, then acute ischaemic algorithm.
3. If indeterminate, then penile/perineal PDUS.
4. If non-ischaemic, then expectant algorithm.

5 · ACUTE ISCHAEMIC PRIAPISM (AUA/SMSNA)

- Irreversible damage begins 4–12 h after presentation, so counsel about ED risk first. Conservative measures are **adjuncts only; never delay first-line treatment**. Oral terbutaline or pseudoephedrine is not recommended.

P3 *Priapism: Diagnosis and Management — ED by duration (SCD series): ≤12 h 0%, 12–24 h 22%, 24–36 h 56%, ≥36 h **100%***

ALGORITHM

1. If <36 h, then aspirate ± irrigate and inject intracavernosal phenylephrine (BP and cardiac monitoring).
2. If that fails, then distal shunt ± tunnelling; tunnel if not done.
3. If still erect, then proximal shunt or penoscrotal decompression; if refractory, prosthesis.
4. If >36 h, then counsel high ED risk with analgesia/follow-up, OR early penile prosthesis.

Aspiration / irrigation

- After a penile block, insert a **19-G needle** at the penoscrotal junction at **3 or 9 o'clock**; one side usually suffices. Aspirate, repeat and irrigate with saline.
- Aspiration alone works in fewer than 1 in 4. Cold (10 °C) irrigation gave 96% vs 60% in a randomised controlled trial (RCT), but **avoid chilled saline in SCD**.

Phenylephrine

- It is a selective α1-agonist with the least cardiac effect.
- Dilute to **100–500 µg/mL** and give **1-mL aliquots every 3–5 min** for up to **1 h** before declaring failure; the maximum is about 1 mg in 1 h (European Association of Urology).
- Toxicity includes hypertension, headache, arrhythmia and **reflex bradycardia**. Monitor, especially with cardiovascular disease or monoamine oxidase (MAO) inhibitors.

Shunts

P3 **P2** *Priapism: Diagnosis and Management; Management of priapism — Winter, Ebbehoj, T-shunt, Al-Ghorab, snake tunnelling (7/8 Hegar), Quackels/Sacher (stagger ≥1 cm), Grayhack/Barry, penoscrotal decompression*

Not in the paper answer:

- AUA: proximal shunts only after a failed distal shunt with tunnelling. Complications include urethral injury or fistula, necrosis, infection and pulmonary embolism.

Early penile prosthesis

- It is used for priapism **>36 h** or failed shunts, because oedematous corpora dilate easily, while delayed surgery meets dense fibrosis.
- Early vs delayed implantation: shortening 3% vs 40% and satisfaction 96% vs 60%.
- Prior distal shunts raise the risk of distal perforation (see Ch 75).

6 · NON-ISCHAEMIC PRIAPISM

*It is **not an emergency**.*

- About 60% resolve with observation, compression and ice.

ALGORITHM

1. Observe and reassess at 4 weeks.
2. If it persists, then PDUS (fistula site, size), counsel, and selectively embolise the internal pudendal/cavernosal artery.
3. If it recurs, then repeat embolisation with a non-resorbable agent.
4. Only after repeated failure, ligate surgically.

Agent	Type	Feature
Autologous clot	Temporary	Children; early recurrence
Gelatin sponge	Temporary (2–6 weeks)	Reabsorbed
Microcoils	Permanent	Theoretical ED risk
Polyvinyl alcohol particles	Permanent	**Preferred for repeat** (AUA)
N-butyl cyanoacrylate	Permanent	Second line for recurrence

- Embolisation succeeds in ≥75%, with 30–40% recurrence. Open ligation causes ED in about 50%.

7 · STUTTERING PRIAPISM — PREVENTION

- An episode ≥4 h is treated as acute ischaemic priapism. Short episodes are managed with fluids, walking and analgesia, and home self-injected phenylephrine for breakthrough.
- No prophylactic drug has robust evidence.

Agent	Dose	Note
PDE5I	Sildenafil 25–50 mg daily; tadalafil 5–10 mg	Most promising; start when flaccid; the only RCT missed its primary endpoint
Pseudoephedrine	60–120 mg at night	Off-label first-line α-agonist
Etilefrine	50–100 mg daily	α1-agonist; Cochrane: insufficient evidence
GnRH agonists, antiandrogens (bicalutamide, flutamide, cyproterone)	Depot / oral	Suppress libido and erections; avoid before puberty is complete
Finasteride	5 mg, tapered to 3 mg then 1 mg	One SCD series
Digoxin	0.25–0.5 mg	Inhibits Na^+/K^+-ATPase
Baclofen	Oral 40 mg at night; intrathecal	Neurogenic/reflex erections

SCD specifics

- First-line aspiration and phenylephrine are the same; give IV hydration and analgesia **concurrently, never instead**. Exchange transfusion is not a primary treatment (AUA); involve haematology and avoid chilled irrigation.

8 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Ateyah 2005	RCT, ICI-induced priapism, irrigation temperature	10 °C saline 96% vs 60% at 37 °C	Cold irrigation (not in SCD)
Burnett 2014	RCT, SCD, sildenafil vs placebo	Primary endpoint missed; open-label benefit	PDE5I prophylaxis plausible, unproven
Zacharakis 2014	Early vs delayed prosthesis	Shortening 3% vs 40%; satisfaction 96% vs 60%	Implant early when ED is inevitable

74 Disorders of Male Orgasm and Ejaculation

1 · SPECTRUM

- Ejaculatory dysfunction (EjD) is the **most common male sexual disorder**. It includes premature ejaculation (PE), delayed ejaculation (DE), anejaculation and retrograde ejaculation (RE). Painful ejaculation, anhedonia and postorgasmic illness syndrome (POIS) are rarer.
- Orgasm is a separate **cortical** event, so ejaculation and orgasm can dissociate.

2 · PHYSIOLOGY OF EJACULATION

- Lumbar interneurons at **L3–L5** form the **spinal ejaculation generator**.

Phase	Events	Mediation
Emission	Vas, seminal vesicles and prostate deliver semen to the posterior urethra	Autonomic, **T10–L2**
Ejection	Bulbospongiosus, ischiocavernosus, levator ani contract rhythmically; **bladder neck closes**; external sphincter relaxes	Somatic **S2–S4 (pudendal)** and a sympathetic reflex
Orgasm	Cerebral processing of pudendal input	Cortical

- **Dopamine (D2) promotes** and **serotonin (5-HT) inhibits** ejaculation.
- 5-HT2C stimulation delays and 5-HT1A shortens ejaculation, so lifelong PE is thought to be **5-HT2C hyposensitivity and/or 5-HT1A hypersensitivity**.

> **KEY POINT**
> Serotonin brakes, dopamine accelerates: PE drugs raise synaptic 5-HT; DE drugs are dopaminergic, noradrenergic or anti-serotonergic.

3 · PREMATURE EJACULATION — DEFINITION AND CLASSIFICATION

Short latency alone is not PE; loss of control and distress are needed.

P1 P3 *Premature Ejaculation — ISSM 2014 three criteria; lifelong/acquired/variable/subjective; AUA 2 min; aetiology, evaluation (IELT, PEDT, IIEF-5), drug table*

Not in the paper answer:

- DSM-5 duration (≥6 months, 75–100% of occasions) and severity bands are in the table below.

	Lifelong	Acquired	Variable	Subjective
Onset	From first encounters, every partner	After previously normal latency	—	—
Latency	≤ ~1 min (mostly 30–60 s)	Markedly reduced, often ≤3 min	Occasionally short	Normal or long
Basis	Neurobiological, genetic	Medical/psychological	Normal variation	Psychological/cultural

	Lifelong	Acquired	Variable	Subjective
Disorder?	Yes	Yes	No	No

- The International Society for Sexual Medicine (ISSM) requires ejaculation within **about 1 min** (lifelong) or a bothersome fall, often to **≤ about 3 min** (acquired). It is the first **evidence-based** definition, for vaginal intercourse only. **Latency alone never suffices**.

	ISSM	DSM-5 (2013)	AUA/SMSNA (2020)
Lifelong latency	≤ ~1 min	≈1 min	≈**2 min** since sexual debut
Acquired	Bothersome fall, often ≤3 min	Specifier only	Marked fall (2–3 min or ≥50%)
Duration/frequency	Always or nearly always	≥6 months, 75–100% of occasions	—
Severity	—	Mild 30–60 s; moderate 15–30 s; severe <15 s	—
Basis	Evidence	Same data	Expert opinion

- The AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America) 2-min line trades specificity for sensitivity.
- Community stopwatch intravaginal ejaculatory latency time (IELT) is positively skewed, median **5.4 min**; ≤1 min is statistically abnormal.

4 · PREMATURE EJACULATION — AETIOLOGY AND EVALUATION

- Lifelong PE affects younger men with few comorbidities; acquired PE affects older men with metabolic syndrome or diabetes.
- **PE with erectile dysfunction (ED)** affects up to half of men with ED, who rush before a partial erection is lost.
- PE is the main sexual symptom in **chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS)**, and it improves when **hyperthyroidism** is treated, though routine thyroid tests are not indicated.
- Examination is desirable in lifelong and **mandatory in acquired PE**, with digital rectal examination (DRE) for prostatitis.

5 · PREMATURE EJACULATION — TREATMENT

ALGORITHM

1. Confirm PE.
2. If it is secondary to ED or another dysfunction, then treat that primary cause.
3. If lifelong, then SSRI ± behavioural therapy (patient preference).
4. If acquired, then treat the cause ± behavioural therapy/SSRI; try withdrawal after 6–8 weeks.
5. If variable or subjective, then reassurance, education, psychotherapy.

- Behavioural therapy is mainly an adjunct.

Drug	Dose / timing	Notes
Dapoxetine	30–60 mg, 1–3 h before	**Only licensed oral PE drug** (>50 countries, not FDA); short half-life; works from the first dose; no PDE5I interaction; many stop it
Paroxetine (daily)	10–40 mg	**Strongest delay**; off-label
Sertraline / fluoxetine / citalopram	50–200 / 20–40 / 20–40 mg daily	Off-label

Drug	Dose / timing	Notes
Clomipramine (a tricyclic antidepressant, TCA)	12.5–50 mg daily or 3–4 h before	Off-label
Topical lidocaine/prilocaine	20–30 min before (spray ≥5 min)	Condom advised; level 1a evidence
Phosphodiesterase type 5 inhibitors (PDE5Is)	30–50 min before	Best with coexisting ED; add-on to SSRI
Tramadol	25–50 mg, 3–4 h before	Low evidence; addiction risk

- **Daily selective serotonin reuptake inhibitors (SSRIs)** act in 5–10 days, fully at **2–3 weeks**.
- Use caution in bipolar disorder, with aspirin or NSAIDs, and under 18 years. **Never stop them abruptly.**
- Meta-analyses show that a PDE5I beats placebo and that a PDE5I with an SSRI beats an SSRI alone.

On-demand dapoxetine	Daily off-label SSRI
Licensed; fewer side effects; little withdrawal; infrequent intercourse	Greater delay; full effect 2–3 weeks; must taper; established relationships

KEY POINT

The real limiter in PE is adherence, not efficacy: most stop SSRIs and topicals (cost, lost spontaneity, "antidepressant" stigma).

6 · DELAYED EJACULATION, ANEJACULATION AND ANORGASMIA

- DE (DSM-5) is a marked delay or absence of ejaculation despite adequate stimulation, on ~75–100% of occasions, with distress. There is **no validated cut-off**, but >25–30 min with distress suggests DE.
- **Anorgasmia** is absent perceived orgasm. DE is mostly acquired and rises with age; its hallmark is normal erections with **low subjective arousal**.

Category	Causes
Afferent	Ageing (penile afferents, Pacinian corpuscles)
Psychogenic	Inhibited ejaculation; performance anxiety; religious orthodoxy; **idiosyncratic vigorous masturbation**
Endocrine	Hypogonadism, **hypothyroidism** (DE; hyperthyroidism causes PE), hyperprolactinaemia
Neurogenic disease	Spinal cord injury (SCI), then diabetic autonomic neuropathy, then multiple sclerosis (MS)
Neurogenic surgery	Radical prostatectomy, rectal surgery, sympathectomy, aortic aneurysm repair, retroperitoneal lymph node dissection (RPLND)
Anatomical (causes RE)	Transurethral resection of prostate (TURP), bladder neck incision
Congenital / infective	Müllerian cyst, Wolffian anomaly, prune belly; urethritis, tuberculosis (TB), schistosomiasis
Drugs	**SSRIs** (~7-fold DE risk), TCAs, antipsychotics, α-methyldopa, thiazides, alcohol

- **Superior hypogastric plexus** injury at RPLND causes dry ejaculation, which **nerve-sparing RPLND** preserves in most men.
- Ask whether masturbation works and whether he has orgasm without semen (RE or obstruction). Examine the vasa and do a DRE; check morning testosterone (T), HbA1c and lipids.

ALGORITHM

1. Orgasm? If never or sometimes, then inhibited orgasm (psychosexual therapy) or ageing (reassure, change technique).
2. If always but no ejaculate, then look for **sperm in post-orgasm urine**; if present, RE.
3. If absent (aspermia), then failure of emission (neurogenic, drug, metabolic) or ejaculatory duct obstruction (EDO).

	Bilateral EDO	**RE**	**Failure of emission**
Antegrade volume	Very low	Low or absent	Absent
pH, fructose	Low (seminal vesicle fluid, ~80% of volume, is missing)	—	—
Post-orgasm urine	No sperm	**Sperm and fructose**	No sperm

Treatment

- Treat the cause, and counsel about anejaculation and sperm retrieval before pelvic surgery. Refer to a psychosexual therapist once organic and drug causes are excluded (AUA).
- Normalise T, change the offending drug and change masturbatory style. Drugs are all off-label, with about 40% success.

Drug	**Action**	**Dose**
Cabergoline	Central D2 agonist	0.5–2 mg every 3 days; **annual echocardiography** (valvular fibrosis)
Bupropion	Dopaminergic	150 mg once or twice daily
Pseudoephedrine	α1 agonist	120 mg 1–2 h before
Reboxetine	Noradrenaline reuptake inhibitor	4–8 mg daily
Cyproheptadine	5-HT antagonist	4–12 mg 3–4 h before (SSRI-induced DE); sedation
Amantadine, buspirone	—	Anecdotal
Intranasal oxytocin	Oxytocinergic	Both RCTs negative

7 · RETROGRADE EJACULATION

P3 *Retrograde ejaculation — bladder neck failure; TURP 42–100%, TUIP 5–45%, diabetic neuropathy; post-ejaculate urine; pseudoephedrine/imipramine doses; urinary sperm retrieval for IUI/IVF/ICSI*

Not in the paper answer:

- Diagnosis: **>5–10 sperm/high-power field** in urine passed after masturbation confirms RE.
- Aquablation preserves ejaculation better than TURP in randomised controlled trials.

8 · SPERM RETRIEVAL IN SCI

- Reflex erection needs the sacral arc, but ejaculation needs thoracolumbar outflow, so it is rare (<5%) with complete upper motor neuron lesions; RE is common.

ALGORITHM

1. **Penile vibratory stimulation (PVS)** is first line, with up to ~70% success; it needs an intact reflex at the T12–S1 level.
2. If PVS fails, then **electroejaculation (EEJ)**: a rectal probe stimulates hypogastric sympathetic efferents; effective and safe.
3. If EEJ fails, then prostate massage, then surgical retrieval (TESE/TESA/MESA/PESA); testicular sperm extraction succeeds in most.

- Lesions above T6 risk **autonomic dysreflexia**: pretreat with **nifedipine** and monitor BP.

9 · OTHER DISORDERS

Disorder	Key points
Painful ejaculation	Prostatitis/CPPS, urethritis, benign prostatic hyperplasia (BPH), seminal vesicle stones, EDO, α-blockers (least with alfuzosin); often none. Surgery (TURP, transurethral resection of ejaculatory ducts (TURED), pudendal neurolysis) weakly supported
Haematospermia	Usually benign (see pointer below). Seminal vesiculoscopy if refractory (see Ch 70); evaluate if >40 y or persistent
Climacturia	Leak at orgasm after radical prostatectomy. Steps: void first, condom, pelvic floor training, constriction ring, male sling
Ejaculatory anhedonia	Orgasm without pleasure; depression, hypogonadism, hyperprolactinaemia, SSRIs
Post-SSRI sexual dysfunction	Genital numbness, low desire, ED, anhedonia persisting months to years
POIS	Myalgia, fatigue, flu-like state within 30 min of orgasm
Orgasmic headache	Dull bilateral ache rising with excitement; sudden, intense at orgasm

P1 *Haematospermia — causes, who needs work-up (≥40 y, persistent, haematuria), PSA/TRUS/MRI, treatment*

75 Surgery for Erectile Dysfunction

1 · PLACE OF SURGERY AND DEVICES

A prosthesis is the last step for erectile dysfunction (ED).

P2 **P3** *Surgical management of erectile dysfunction and penile prosthesis; Types of penile prosthesis — last-line but most effective (satisfaction 80–90%; repair 15% at 5 y, 30% at 10 y); Lue goal-directed approach; malleable vs 2-piece vs 3-piece; device names, expansion, coatings; consent; pre-op preparation; revascularisation*

Not in the paper answer:

- Coloplast cylinders are polyurethane (Bioflex), so keep cautery **<35 W**; Boston and Rigicon silicone cylinders tolerate cautery.

2 · SELECTION AND COUNSELLING

- **Consent is expectation management**: the device restores rigidity, not the former erection.
- The commonest disappointment is a **shorter erection**, because the pseudocapsule limits expansion (worse in Peyronie disease). Document it.
- With **phimosis**, circumcise first and implant about 6 weeks later.
- Raised glucose on the day needs intravenous (IV) insulin. Nasal methicillin-resistant Staphylococcus aureus (MRSA) carriers get chlorhexidine bathing and mupirocin 2% for 5 days.

3 · SURGICAL APPROACHES

Approach	Advantages	Cautions
Penoscrotal	Majority choice; secure subdartos pump; good crural exposure	Urethra at risk, so Foley first; reservoir via inguinal canal
Infrapubic (Scott)	Reservoir under vision	Dorsal neurovascular bundle: stay sutures at 10 and 2 o'clock; patient pulls pump down daily
Subcoronal	Rods; allows straightening	Circumcise at the same time, or lymphoedema follows
Ventral penile	Rods without bending; foreskin kept	—

4 · OPERATIVE TECHNIQUE (THREE-PIECE)

ALGORITHM

1. Catheter; expose the tunica.
2. Two 2-0 polydioxanone (PDS) stay sutures per corpus; longitudinal corporotomy.
3. Dilate proximal, then distal (Metzenbaum, Hegar/Brooks 7→13 mm).
4. Goalpost test; measure (Furlow).
5. Cylinders in; test inflation 1. Close corporotomies; test inflation 2.
6. Reservoir, then pump; connect; test inflation 3.
7. Irrigate and close.

- Dilators point **dorsolaterally**, away from urethra and septum. The **goalpost test** (equal height and symmetry) detects crossover.
- The cylinder is equal to or 1 cm shorter than measured, because oversizing causes buckling and S-deformity; rear tip extenders adjust length.
- The pump sits in a dependent subdartos pouch, anterolateral to the testis.

Reservoir placement

- The **Retzius** space is entered with the bladder empty; the iliac vessels lie about 1 inch lateral.
- In a hostile pelvis (radiation, cystectomy, robotic prostatectomy, transplant) use an **ectopic** site, made safe by the **lockout valve**.
- **Submuscular suprafascial** is the commonest ectopic site. Fill it nearly full, because underfilling leads to migration.

5 · INTRA-OPERATIVE PROBLEMS

- **Fibrotic corpora** follow priapism or explant (pelvic radiation does not add difficulty). Use cavernotomes or scar excision and **narrow cylinders**, always two.
- **Crossover** gives a lopsided erection and unequal measurements. Leave a large dilator in the common cavity and redilate the other side laterally; the Foley should lie midline between cylinders.
- **Crural perforation** (one dilator deeper, lengths differ by >1 cm) is repaired with a **suture sling** (2-0 Prolene through the rear tip, over the corporotomy).
- **Urethral perforation** at the corporotomy is closed and the operation proceeds. If distal (fossa navicularis, irrigant at the meatus), **abandon that cylinder**, place the contralateral one, leave no catheter and replace at 5–6 months.
- A **hypermobile glans** drooping ventrally is the supersonic transporter (**SST**) deformity. It is fixed with glans sutures (3-0 non-absorbable).
- **Residual curvature** is treated with high-rigidity cylinders (AMS 700 CX, Titan, Infla 10 X) and **modelling** (clamp tubing, bend 90° away for 90 s), then plication or incision with or without graft. Less than 20° is acceptable (see Ch 76).

KEY POINT

No catheter after distal urethral perforation: the urinary stream draws fluid into the urethra (Bernoulli), so urine does not enter the defect. Wait 5–6 months for scar to obliterate the false tract. Most intra-operative errors are **prevented by instrument direction** and **detected by symmetry** (goalpost test, midline Foley).

KEY POINT

After robotic prostatectomy the peritoneum is not replaced and bowel lies near the internal ring, so a blind inguinal Retzius reservoir can injure bowel. Choose an ectopic site or place it under vision.

6 · COMPLICATIONS

Infection usually costs the whole device.

Infection (1–3%)

- Risks include hyperglycaemia, revision, MRSA and scarred corpora.
- American Urological Association (AUA) prophylaxis is IV vancomycin, or a 3rd-generation cephalosporin plus gentamicin, 1 h before incision and for ≤24 h. Extended oral courses are unproven; in high-risk men add an antifungal, with or without piperacillin–tazobactam.
- Coatings (InhibiZone rifampin–minocycline; hydrophilic Titan/Infla) halve infection, and a **no-touch** technique gives <0.5%.
- Fever and erythema suggest infection; pus, an exposed device, progressive pain or a **pump fixed to the scrotal wall** are definitive. **Remove ALL components**, as retained parts harbour biofilm.

Option	Technique	Caveats
Explant, delayed replacement	Antibiotic irrigation drains for 3 days; reimplant at 2–6 months	Fibrosis, loss of several cm; vacuum erection device (VED) meanwhile
Mulcahy salvage	Remove all, antiseptic washes, new drapes, immediate new device	~84%; fails with onset <2 months, aggressive organisms, purulence with cellulitis
Delayed salvage	IV vancomycin and gentamicin, salvage at 3 days	Aggressive infection
Malleable salvage	Rods instead	Quick; convert later

- Washes use dilute povidone-iodine (0.35–3.5%) for 3 min, then saline or antibiotic; hydrogen peroxide is dropped as cytotoxic.
- An uninfected revision carries ~10% infection, falling to ~3% with antiseptic washout of all cavities.

Mechanical failure

- This is the **commonest complication**: leak, pump malfunction, autoinflation, kinking or cylinder aneurysm.
- If the device is >3 years old, replace **all** parts; check any retained part with the **ohmmeter test**. A rod that loses rigidity has a core fracture and is replaced.

Extrusion and erosion

Feature	Extrusion	Erosion
Position	Tip through tunica, still under skin/glans	Through skin or into urethra
Cause	Forceful dilation, oversizing, aggressive use	Infection, crossover, catheterisation; rods in insensate men (spinal cord injury)
Management	**Distal corporoplasty** (new dorsomedial plane behind pseudocapsule)	Remove all, washout, contralateral spacer, replace at 5–6 months

- **Reservoir erosion** into the bladder needs reservoir exchange and repair; into bowel, remove the whole device.

- **Penile necrosis** is linked to diabetes, vascular disease, smoking, prior implant and radiation. Avoid circumferential incisions and tight wraps, and remove cylinders promptly if it occurs.
- An early basal bulge is corporotomy disruption (repair); a late one is cylinder aneurysm (replace).

7 · SPECIAL SITUATIONS

- After **priapism** >36 h, an **immediate implant** preserves length and relieves pain; avoid distally expanding cylinders (see Ch 73).
- In **Peyronie disease** length loss is greater, so use non-expanding high-rigidity cylinders with modelling.
- Transplant and HIV-positive men (adequate CD4) have no excess infection.

8 · POST-OPERATIVE CARE

- It is day surgery. Leave cylinders **partially inflated** so the pseudocapsule forms large, remove the drain within 24 h and keep the penis upward for 4–6 weeks.
- Activation is at **4–6 weeks**, then the device is cycled daily. Dissatisfaction is mainly about length and rigidity.

9 · PENILE REVASCULARISATION †

- †Only young men (<55) with a focal post-traumatic pudendal occlusion qualify (inferior epigastric to dorsal artery). It is not recommended for generalised atherosclerosis or venous leak.

76 Diagnosis and Management of Peyronie Disease

1 · DEFINITION

- **PD** (American Urological Association, AUA 2015) is an acquired penile abnormality with **fibrosis of the tunica albuginea**, with or without pain, deformity, erectile dysfunction (ED) and distress.

> **KEY POINT**
> Myths corrected: PD is **not rare** (3–20%; higher in diabetes and ED). **Spontaneous resolution is rare** and up to half worsen untreated, so "wait 6–12 months" is a myth. **Calcification is not a marker of chronic disease**.

2 · ANATOMY AND PATHOPHYSIOLOGY

- The outer longitudinal tunica layer is **absent at 5–7 o'clock**, so the dorsum buckles; **60–70% of plaques are dorsal** with septal involvement.
- **Microtrauma** from bending during intercourse causes **delamination at the septal insertions**.
- **Transforming growth factor-β1 (TGF-β1)** is central. Collagen becomes disorganised and elastin decreases, so the penis **curves toward the plaque**.
- A **susceptible host** is needed, and PD is associated with **Dupuytren and Ledderhose** disease; only 16–40% recall trauma.

3 · CLINICAL FEATURES AND PHASES

- Features are pain, deformity (curvature, indentation, **hourglass**, hinge, shortening), plaque and ED. Hourglass or hinge gives an **unstable penis**.
- Kelami grading: ≤30° mild, 31–60° moderate, >60° severe. Ventral and lateral curves are harder to manage than dorsal.

Feature	Active (acute)	Stable (chronic)
Pain	Present (flaccid, erection, intercourse)	Usually resolved
Deformity	Progressive	No progression
Aim	Prevent progression; non-steroidal anti-inflammatory drugs (NSAIDs) for pain; traction	Correct deformity (injection, surgery)
Surgery	Not appropriate	When **≥12 months from onset and stable ≥6 months**

- The differential includes congenital curvature, chordee, post-traumatic fibrosis and tumours (epithelioid sarcoma mimics PD).

4 · EVALUATION

- The **key question** is: "If straight with current rigidity, would it be adequate for penetration?" If **no**, offer a prosthesis with straightening; the answer also predicts post-operative erectile function.
- Measure **stretched penile length (SPL)** from pubis to corona dorsally; this is essential, as patients overestimate loss.
- Extensive **calcification** on ultrasound predicts poor intralesional response and more surgery.

Levine calcification grade	Size	Surgery likelihood
1	<0.3 cm	Not increased
2	0.3–1.5 cm	Not increased
3	>1.5 cm, or multiple ≥1 cm	Increased

- **Intracavernosal injection (ICI) with or without penile duplex Doppler ultrasonography (PDDUS)** is required by the AUA **before any invasive treatment**. It measures curvature and gives peak systolic velocity, end-diastolic velocity and resistive index (RI).

5 · NON-SURGICAL MANAGEMENT

P1 *Management of Peyronie's disease — phases, evaluation, non-surgical options table, surgical algorithm*

Not in the paper answer:

- **Oral**: the AUA "should not offer" any except **NSAIDs for pain**; vitamin E, tamoxifen, colchicine and carnitine give no benefit.
- **Extracorporeal shock wave therapy (ESWT)** helps pain, not curvature; radiotherapy is harmful.

Collagenase Clostridium histolyticum (CCH, Xiaflex)

- It is the only PD drug approved by the Food and Drug Administration (FDA, 2013). It degrades collagen **types I and III** and spares elastin, vessels and nerves.
- **Protocol (IMPRESS)**: 2 injections of **0.58 mg** 24–72 h apart, then **model** the penis 24–72 h after the second. Give up to **4 cycles, 6 weeks apart** (max 8 injections). Modelling is essential.
- **AUA criteria** are stable disease, curvature **>30° and <90°** and intact erections (with or without medication). IMPRESS excluded ventral plaques, extensive calcification and disease <12 months.
- Response is poor with calcification, curve ≥60°, severe indentation or hinge.
- Bruising and pain are common; the serious risk is **corporal rupture** (0–4.9%), so there is no intercourse until modelling ends.

Other intralesional agents

Agent	Mechanism	Evidence	AUA
Verapamil	Calcium channel blocker (CCB): reduces fibroblasts and extracellular matrix, increases collagenase	Randomised trials conflicting; stabilises; inferior to CCH; cheap	May offer

Agent	Mechanism	Evidence	AUA
Interferon α-2b	Reduces fibroblasts and collagen, increases collagenase	Randomised controlled trial (RCT) vs saline: curvature reduced 27% vs 9%; flu-like symptoms (pre-treat with NSAIDs)	May administer
Nicardipine	CCB	Reduces pain and plaque; no curvature benefit	—

- Verapamil is given as 10 mg diluted (10–20 mL).

Mechanical therapy

- **Penile traction therapy (PTT)** in the acute phase reduces curvature, pain and the need for surgery, and gains length; it needs at least 3 h/day.
- The AUA makes no recommendation for traction, vacuum or combinations.

6 · SURGERY

Indications

- Disease is **≥12 months from onset, with deformity stable ≥6 months**.
- Deformity or poor rigidity prevents intercourse, and conservative therapy has failed.

Consent

- The goal is a **functionally straight** penis (≤20°; European Association of Urology, EAU, ≤15°). Residual or recurrent curvature occurs in up to 16%.
- **Length loss** follows every procedure (plication more than grafting); 70–80% have already lost length, so document SPL.
- **New ED** is commoner after grafting than plication (~30% if pre-op rigidity is suboptimal); pre-op erection quality predicts it best.

Choice of procedure — algorithm

ALGORITHM

1. Ask: is the erection adequate (with or without a PDE5I)?
2. If **no**, place an inflatable prosthesis with modelling. If the curve persists, incise with or without a graft (graft if defect >2 cm).
3. If **yes**, do PDDUS. If the curve is simple and <60–70°, with no hourglass or hinge and acceptable length loss, **plicate** (if length loss is unacceptable, graft).
4. If **yes** but the curve is complex/multiplanar >60–70°, or there is severe hourglass, hinge or a short penis (<9 cm stretched), do **plaque incision/partial excision with graft** if the ED risk is accepted (otherwise prosthesis).

Plication (shortening)

- It shortens the **convex side opposite the plaque**. It is simple, with the least new ED, but does not fix hourglass, hinge or shortening.

Technique	Method	Point
Nesbit (1965)	Transverse elliptical full-thickness excision opposite the curve, closed transversely	Original; for a ventral curve, lift Buck fascia and take dorsal wedges
Yachia	Vertical 0.5–1.5 cm incision closed transversely (Heineke–Mikulicz)	No tunica removed; long incisions narrow the shaft
Essed–Schroeder	Non-absorbable figure-of-eight, buried knots	First incisionless plication
16-dot	Permanent extended Lembert, 4 dots per plication	Popular; no incision

Technique	Method	Point
Tunica albuginea plication (TAP, Levine)	Paired partial-thickness transverse incisions (outer layer only); buried permanent suture and 3-0 polydioxanone (PDS)	Cavernosal tissue undisturbed

Plaque incision/partial excision and graft (lengthening)

- It suits complex deformity with **strong erections** and no veno-occlusive dysfunction, and restores calibre and 0.5–3 cm of length.
- ED predictors are age >55, veno-occlusive dysfunction (RI <0.80), a large graft, a ventral curve and a curve >60°.
- **Key steps**: elevate Buck fascia with the neurovascular bundle, make an H/double-Y incision **crossing the septum**, and sew in the graft (pericardium or small intestinal submucosa) with running 4-0 PDS.

Prosthesis with straightening

- PD with PDE5I-refractory ED is treated in one operation; an **inflatable** device is preferred.
- **Manual modelling** (Wilson–Delk): inflate fully, clamp the tubing and bend opposite the curve for 60–90 s, with the hand on the shaft, not the glans. It straightens 86–100%.
- Residual curvature >30° needs a relaxing incision, with a graft if the defect is >2 cm. Length loss is the commonest complaint (see Ch 75).

7 · LANDMARK TRIALS

Trial	Design	Result	Take-home
IMPRESS I and II	Phase III RCT, CCH vs placebo 2:1, curvature 30–90°	Curvature reduced 34% (17°) vs 18% (9.3°); bother improved	Basis of FDA approval; modelling essential
Interferon α-2b multicentre RCT	Intralesional interferon (IFN) vs saline	Curvature reduced 27% vs 9%	First RCT of intralesional therapy

★ Putting it together: andrology decisions

A · INFERTILE MAN: WORK-UP

ALGORITHM

1. **History and examination**: testosterone use, mumps, cryptorchidism; testis volume, vasa, varicocele (see Ch 69).
2. **Two semen analyses** (abstinence 2–7 days). Low volume with acid pH suggests duct obstruction or congenital bilateral absent vas (CBAVD).
3. **Hormones**: follicle-stimulating hormone (FSH) and testosterone; if testosterone is low, add luteinising hormone (LH) and prolactin.
4. **Genetics** (azoospermia, severe oligozoospermia): karyotype, Y microdeletions; CFTR in CBAVD.

B · AZOOSPERMIA: OBSTRUCTIVE OR NOT

	Obstructive (OA)	Non-obstructive (NOA)
Testis	Normal volume	Small, soft
FSH	Normal (≤7.6 IU/L)	Raised (>7.6 IU/L)
Clues	Absent vasa, low volume, vasectomy or hernia repair	Klinefelter, Y microdeletion, cryptorchidism, chemotherapy

	Obstructive (OA)	Non-obstructive (NOA)
Treatment	Vasovasostomy (VV) or vasoepididymostomy (VE), resection of ejaculatory ducts, or sperm retrieval for ICSI (see Ch 70)	Micro-TESE with ICSI; fix a clinical varicocele first; no retrieval if complete AZFa/b deletion

KEY POINT

† Reversal suits a younger partner wanting >1 child; ICSI suits an older partner or female factor. Copious vas fluid or sperm means VV; toothpaste-like fluid or none means VE (see Ch 70).

C · ERECTILE DYSFUNCTION LADDER

ALGORITHM

1. **Assess:** IIEF-5 score, HbA1c, lipids, morning testosterone. ED is a cardiovascular sentinel, so stratify cardiac risk (see Ch 72).
2. **Modify** risk factors; treating hypogonadism rescues PDE5 inhibitor (PDE5I) failures.
3. **Shared decision:** PDE5I (never with nitrates), vacuum, intraurethral alprostadil or intracavernosal injection.
4. **Penile prosthesis** (inflatable three-piece) if others fail; highest satisfaction. (see Ch 75).
5. **Young, focal post-traumatic arterial occlusion** → revascularisation.

D · PRIAPISM: ISCHAEMIC VS NON-ISCHAEMIC

	Ischaemic (emergency)	Non-ischaemic (high-flow)
Clinical	Rigid, painful	Semi-rigid, painless; after perineal trauma
Cavernous blood gas	pO_2 <30, pCO_2 >60 mm Hg, pH <7.25	Like arterial blood: pO_2 >90, pCO_2 <40
Doppler	Little or no flow	High flow, fistula or pseudoaneurysm
Treatment	Aspirate with or without irrigation, then phenylephrine (100–500 µg every 3–5 min, up to 1 h), then distal shunt with or without tunnelling, then proximal shunt. >36 h or refractory: early prosthesis	Observe; selective embolisation if persistent

KEY POINT

Sickle cell: treat as above without waiting for transfusion. Stuttering priapism: daily PDE5I, nightly pseudoephedrine, or hormonal suppression after puberty (see Ch 73).

E · PEYRONIE DISEASE

ALGORITHM

1. **Active**: analgesia, traction. Collagenase (CCH): stable dorsal/lateral 30–90°, intact erections; else verapamil, interferon α-2b.
2. **Surgery**: ≥12 months from onset, stable ≥6 months (see Ch 76).
3. If **erections are good**: simple <60–70°, no hourglass/hinge, adequate length → plication; complex, hourglass/hinge or short → incision/partial excision + graft (more ED).
4. If **erections are poor** despite PDE5I → inflatable prosthesis with modelling.

F · PREMATURE EJACULATION (PE)

KEY POINT

Diagnosis needs short latency (lifelong ≈1 min or less), poor control and distress. Treat ED first. Options: stop–start or squeeze, lidocaine–prilocaine spray, on-demand dapoxetine (only licensed oral drug outside the US), daily SSRI (paroxetine best), tramadol (off-label, addiction risk) (see Ch 74).